

Bayesian structural time-series analysis of temperature extremes

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Abstract

Environmental extremes are moving targets: in a changing climate, their exceedance probabilities and return periods must be tracked through time. We embed generalized extreme-value (GEV) block extrema in a Bayesian structural time-series model and, in parallel, use Gaussian observation models for block means. The application follows both ends of the temperature distribution alongside its averages, with a distinct level, slope and seasonal cycle for each of six summaries. Hierarchical normal shrinkage lets the summaries jointly inform the strength of evolution while retaining their own trajectories. Half-normal hyperpriors are calibrated through the orders of magnitude of long-term temperature changes, and joint prior simulations separate their implications from those of the initial-state and observation priors. Weak climatic evolution is difficult to distinguish from weather variability and can couple latent paths tightly to their innovation scales, a common difficulty in state-space inference. A non-centred representation stabilizes learning near the deterministic limit. For GEV margins, a local Laplace approximation yields complete-path proposals by Gaussian smoothing; an independence Metropolis–Hastings correction preserves the exact posterior, with 95–99% acceptance in this application. Applied to a 135-year Belgian daily temperature record, the fitted rates of warming generally rise from the early to the recent period. Daytime trajectories change little around mid-century before renewed warming after 1970, while nighttime trajectories rise more steadily. The fitted probability that a summer contains a day above 35 °C rises from 3.6% in the earliest 30 years to 15.9% in the most recent. Yet a refit before the record summer of 2019 still regarded its record-shattering temperature as highly unlikely. The framework lets us track evolving risk while learning the strength of persistent change from the data.

Keywords: nonstationary extremes; structural time series; Bayesian shrinkage; seasonal temperature; Bayesian computation; environmental risk.

1 Introduction

An event is exceptional only relative to a distribution. The warning that *stationarity is dead* (Milly et al., 2008) is especially consequential for heat: a threshold that once seemed remote can become increasingly likely as temperatures rise (Fischer et al., 2021). Exceedance probabilities and return levels must therefore be treated as evolving quantities, with uncertainty in that evolution carried into risk assessment (Cooley, 2012; Slater et al., 2021).

Average temperatures, hot extremes and cold extremes need not share the same history. Their changes can differ across seasons and between day and night (Donat and Alexander, 2012; Franzke, 2015; McKinnon et al., 2016; Dunn et al., 2019). Local records also reflect atmospheric circulation and cloudiness, variations in surface solar radiation, and changes in the surrounding landscape, including urbanization (IPCC, 2021; Wild, 2009; Hamdi et al., 2009; Hamdi and Van de Vyver, 2011). These differences matter for risk: a shift in background conditions can make a formerly rare temperature much more frequent, while changes in dispersion or tail behaviour can modify that effect (Huybers et al., 2014). We therefore study averages alongside both hot and cold extremes and translate their evolution into event probabilities.

Extreme-value theory (EVT) supplies models for block maxima and minima and supports extrapolation beyond routinely observed values (Fisher and Tippett, 1928; Coles, 2001). Nonstationarity can be addressed through detrending, covariate-dependent distributional parameters, local likelihoods, smooth terms or changepoints (Mentaschi et al., 2016; Katz, 2013; Robin and Ribes, 2020; Davison and Ramesh, 2000; Beaulieu et al., 2012; Youngman, 2022). These approaches differ in how they represent and regularize change. A fixed linear trend may be too rigid for a record spanning more than a century, while a highly flexible curve can follow short-lived weather fluctuations. The challenge is to describe persistent distributional change while distinguishing it from observation variability.

Structural time-series, or unobserved-components, models provide an interpretable way to make this distinction. Established in econometrics and Bayesian dynamic modelling, they represent a series through latent components with explicit temporal roles (Harvey, 1990; West and Harrison, 2006; Durbin and Koopman, 2012). A level tracks background conditions, a slope describes their local rate of change, and a seasonal component captures the annual cycle. Gaussian transition equations allow these components to evolve without requiring Gaussian observations. Linking the latent location of a block directly to a generalized extreme value (GEV) likelihood combines this structural interpretation with an extreme-value observation model, connecting historical reconstruction and forecasting.

An early dynamic GEV (DGEV) formulation by Gaetan and Grigoletto (2004) established this connection through *semiparametric smoothing*. Gaussian random-walk priors allowed GEV parameters to evolve without prescribing a fixed temporal form, with innovation variances controlling their smoothness. Their applications used selected variance values, assessed through posterior predictive comparisons. Similarly, Fearnhead et al. (2010) selected a trend-evolution variance through particle estimates of the likelihood before conducting Bayesian inference for other parameters. These analyses demonstrated the value of flexible latent trajectories while conditioning on a chosen smoothing strength. Further details of their variance specifications are given in [appendix S1.3](#).

Learning that strength jointly with the latent trajectory presents a further challenge. Huerta and Sansó (2007) introduced an auxiliary Gaussian layer between the structural state and the GEV location, facilitating computation through conditional forward filtering and backward sampling. Their variance specification combined inverse-gamma priors with discount factors. This additional location noise is unnecessary for a direct DGEV formulation and introduces a variance component whose contribution may be difficult to distinguish from persistent state evolution and GEV observation dispersion, complicating interpretation. Wei and Huerta (2017) instead linked the GEV location directly to a random walk, assigning its innovation variance a lognormal prior within their centred MCMC benchmark. They reported convergence difficulties with shorter runs and used very long chains. For their particle-filter comparisons, they fixed the variance at a preliminary MCMC posterior estimate after encountering instability when estimating it jointly with the other parameters. Their simulation example used an innovation scale comparable to the GEV scale, leaving performance under much weaker evolution a separate question.

This weak-evolution regime is particularly relevant to gradual climatic change: small state innovations can accumulate over decades while remaining difficult to distinguish from observation variability, which can be substantial for extremes. Fixing their variances omits uncertainty about smoothing strength, whereas estimating them requires careful prior specification. Both inverse-gamma and lognormal priors placed directly on process variances have densities that vanish at zero; their hyperparameters determine how much prior probability is assigned to nearly deterministic evolution. Nominally "vague" specifications can therefore become influential in this regime (Gelman, 2006). A separate computational difficulty arises because small variances tightly constrain latent increments, so alternating updates of centred paths and variances can mix very slowly (Papaspiliopoulos et al., 2007; Fearnhead and Sherlock, 2026). Learning weak evolution consequently requires attention to both regularization and parameterization, supported

by sensitivity and convergence checks.

We address these issues in a Bayesian structural time-series framework for environmental extremes. A non-centred representation expresses each trajectory through a deterministic trend and seasonal cycle plus standardized stochastic paths multiplied by signed innovation coefficients (Frühwirth-Schnatter and Wagner, 2010). Normal shrinkage priors regularize these coefficients, with one learned prior scale for each type of innovation shared across related responses. This pooling allows the series to inform how much evolution is supported without imposing a common trajectory. It is particularly useful when a single record leaves the allocation of change between level and slope sensitive to its prior calibration. Half-normal hyperpriors express uncertainty about the shared scales. Their calibration concerns the orders of magnitude of plausible evolution over 30 years, rather than an attempt to make the priors as diffuse as possible. We assess these choices through the temperature paths and observations they generate, following the prior-predictive approach of Gabry et al. (2019); the implications of a prior depend on the state dynamics and observation model in which it is used (Gelman et al., 2017). Comparisons with unpooled fits and alternative hyperprior families assess whether borrowing information stabilizes the scientific conclusions.

For GEV observations, a local Laplace approximation supplies a complete-path proposal through Gaussian simulation smoothing. An independence Metropolis–Hastings correction preserves the specified posterior target: the approximation changes the proposal distribution without introducing an additional observation-noise layer. In this application, 95–99% of proposed GEV paths are accepted. This exploits the latent Gaussian structure and provides an alternative to general particle MCMC updates (Andrieu et al., 2010; Lindsten et al., 2014), for the individual extreme-value series considered here.

We apply the method to four seasonal extrema from a 135-year Belgian daily temperature record and analyse the two corresponding seasonal means in parallel using Gaussian structural models. These six summaries follow both ends of the daytime and nighttime temperature distributions alongside their averages. Each has its own evolving trajectory and observation parameters; sharing is confined to the three scales governing the shrinkage of state innovations. We ask how average, hot and cold conditions have changed, how the estimated warming rates have evolved, and what these changes imply for threshold risks and forecasts. The following sections present the data, model and computation, followed by model assessment, historical changes and evolving risks. The discussion considers limitations and extensions to other environmental extremes.

2 Data and exploratory analysis

We use almost 135 years of daily maximum (TX) and minimum (TN) temperatures recorded at the Royal Meteorological Institute of Belgium station in Uccle, Brussels. The homogenized series runs from January 1892 through August 2026; Bertrand et al. (2018) describe its treatment of changes in observing conditions, including changes to the reference thermometer shelter.

We group the daily observations into meteorological seasons: December–February (DJF), March–May (MAM), June–August (JJA) and September–November (SON). We label each DJF block by the year in which February falls. Because the record starts in January 1892, its first winter is incomplete and is omitted. This leaves $T = 538$ complete blocks, from MAM 1892 through JJA 2026: 135 MAM and JJA blocks, and 134 DJF and SON blocks. Let B_t denote the days in block t . The six summaries are defined in table 1.

The suffixes m , x and n denote the block mean, maximum and minimum, following the general ETCCDI notation (ETCCDI, 2024). Seasonal aggregation retains the annual cycle while supplying around 90 daily values per block. The sensitivity analysis in appendix S2 uses the same seasonal observations in every fit, so changes reflect the prior or model specification rather than the aggregation window.

Table 1: Seasonal temperature summaries.

Response	Definition
TXm	$ B_t ^{-1} \sum_{d \in B_t} \text{TX}_d$
TNm	$ B_t ^{-1} \sum_{d \in B_t} \text{TN}_d$
TXx	$\max_{d \in B_t} \text{TX}_d$
TNx	$\max_{d \in B_t} \text{TN}_d$
TXn	$\min_{d \in B_t} \text{TX}_d$
TNn	$\min_{d \in B_t} \text{TN}_d$

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Figure 1: Seasonal temperature summaries at Uccle, 1892–2026. Left: observed summaries, with each colour tracing one season across years. Right: LOESS smoothers of all six summaries after centring each response within season over the full record. All curves use an 18% span and are meant to be purely descriptive.

The left panel of [figure 1](#) shows substantial year-to-year scatter around the long-term changes. Winter is especially variable for the two means and the cold-side extrema, whereas the hottest daytime extreme (TXx) also varies strongly in autumn. The dispersion pattern therefore depends on both season and response.

The overlaid smoothers in the right panel make the different histories easier to see. It suggests that the three daytime summaries seem to flatten or even dip between roughly 1950 and the late 1970s, and then rise strongly; seemingly faster than before the dip. The nighttime summaries on the other hand show no comparably clear cooling interval and rise more steadily overall. Widespread reductions in surface solar radiation during the 1950s–1980s, followed by partial brightening, most likely provide context for such a day–night contrast ([Wild, 2009](#)).

These observations motivate response-specific evolving levels and slopes, observation scales that differ by season, and regularization that separates persistent change from short-term fluctuations. The six summaries provide complementary views of the same daily temperature record.

3 Methods

For summary $j = 1, \dots, 6$, write $\mathbf{Y}_j = (Y_{j,1}, \dots, Y_{j,T})$, $\mathcal{X}_j = \mathbf{x}_{j,0:T}$, and let Θ_j contain its static parameters. With $\boldsymbol{\tau} = (\tau_\alpha, \tau_\beta, \tau_\gamma)$, the hierarchical state-space model factorizes as ([Särkkä, 2013](#))

$$\begin{aligned}
 p(\boldsymbol{\tau}, \Theta, \mathcal{X}, \mathbf{Y}) &= p(\boldsymbol{\tau}) \prod_{j=1}^6 p(\Theta_j \mid \boldsymbol{\tau}) p(\mathbf{x}_{j,0} \mid \Theta_j) \\
 &\times \prod_{j=1}^6 \prod_{t=1}^T p(\mathbf{x}_{j,t} \mid \mathbf{x}_{j,t-1}, \Theta_j) g_{j,t}(Y_{j,t} \mid \mathbf{x}_{j,t}, \Theta_j).
 \end{aligned} \tag{1}$$

We use a product of the conditional observation likelihoods. Responses borrow information through the learned prior scales, while retaining distinct state paths and observation parameters. Conditional on $\boldsymbol{\tau}$, their parameter and state priors factorize across responses. The following subsections specify the states, observations, priors and computation; the supplement gives the state-space matrices and full target.

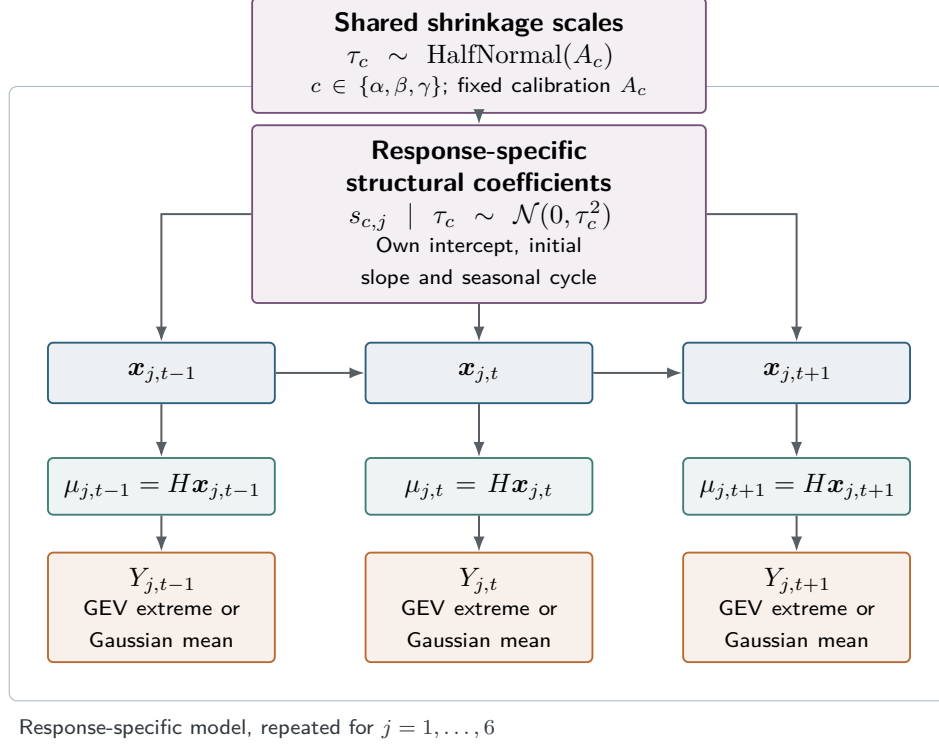


Figure 2: Structural model with shared shrinkage. Each response has its own level, slope and seasonal state, linked to a GEV observation distribution for extrema or a Gaussian distribution for means. Three scales outside the response plate govern the normal priors on the signed innovation coefficients and are learned from all six summaries. Their half-normal hyperpriors have fixed calibration scales. The initial-slope prior remains separately calibrated. Observation scales and GEV shapes are omitted for clarity.

3.1 State transition

Each response has its own level $\alpha_{j,t}$, slope $\beta_{j,t}$ and dummy-seasonal component $\gamma_{j,t}$. For seasonal period $p = 4$, define

$$\mathbf{x}_{j,t} = (\alpha_{j,t}, \beta_{j,t}, \gamma_{j,t}, \gamma_{j,t-1}, \dots, \gamma_{j,t-p+2})^\top, \quad H = (1, 0, 1, 0, \dots, 0). \quad (2)$$

The state determines the evolving location through

$$\mu_{j,t} = H\mathbf{x}_{j,t} = \alpha_{j,t} + \gamma_{j,t}. \quad (3)$$

Its components evolve according to

$$\alpha_{j,t} = \alpha_{j,t-1} + \beta_{j,t-1} + s_{\alpha,j}\varepsilon_{\alpha,j,t}, \quad (4a)$$

$$\beta_{j,t} = \beta_{j,t-1} + s_{\beta,j}\varepsilon_{\beta,j,t}, \quad (4b)$$

$$\gamma_{j,t} = -\sum_{\ell=1}^{p-1} \gamma_{j,t-\ell} + s_{\gamma,j}\varepsilon_{\gamma,j,t}. \quad (4c)$$

The innovations are independent standard normal variables under the state prior, and $q_{c,j} = s_{c,j}^2$ is the process variance of component c .

This local-linear-trend model with stochastic dummy seasonality is a standard Bayesian unobserved-components construction (Harvey, 1990; West and Harrison, 2006; Petris et al., 2009; Scott and Varian, 2014). The level describes the background location, the slope its local rate of change, and the seasonal component the departures associated with DJF, MAM, JJA and SON. Small innovations allow the trajectory to bend gradually without prespecifying changepoints or

a spline basis, while keeping observation variability separate. As all process variances approach zero, the location reduces to a deterministic linear trend plus a fixed seasonal cycle. Interventions, additional cycles and external regressors could be included in the same framework; we use this parsimonious specification as a starting point for nonstationary extremes.

3.1.1 Non-centred representation

Following [Frühwirth-Schnatter and Wagner \(2010\)](#), we separate each stochastic path from its physical amplitude. The paths have standardized innovations, while signed coefficients determine their contributions to the location. This reduces the tight coupling between paths and process scales near the deterministic limit and makes continuous shrinkage on the amplitudes natural ([section 3.3.1](#)).

Let $\tilde{\alpha}_{j,t}$ be a standardized level random walk, $\tilde{A}_{j,t}$ the integrated contribution of a standardized slope random walk, and $\tilde{\gamma}_{j,t}$ a standardized dummy-seasonal process, all initialized at zero. Write $\gamma_{j,s}^{(0)}$ for the initial seasonal cycle, with $\sum_{s=1}^p \gamma_{j,s}^{(0)} = 0$. Then

$$\mu_{j,t} = \alpha_{j,0} + t\beta_{j,0} + \gamma_{j,s(t)}^{(0)} + s_{\alpha,j}\tilde{\alpha}_{j,t} + s_{\beta,j}\tilde{A}_{j,t} + s_{\gamma,j}\tilde{\gamma}_{j,t}. \quad (5)$$

The first three terms give a fixed intercept, linear trend and seasonal cycle; the others allow persistent departures from them. The coefficients $s_{c,j}$ may have either sign, while $|s_{c,j}|$ is the corresponding innovation standard deviation and $q_{c,j} = s_{c,j}^2$ its process variance. Conditional on the standardized paths, the signed scales act as regression coefficients. As a scale approaches zero, its path remains well defined and simply contributes less to the location. The reparameterization preserves the state-transition model and its conditional observation likelihood. The standardized recursions and their equivalence to [equation \(4\)](#) are given in [appendix S1.1](#).

3.2 Observation distributions

The state determines the location parameter $\mu_{j,t}$ of the observation distribution. For extrema, this extends the structural time-series approach beyond the Gaussian and exponential-family settings ([Helske, 2017](#)). In parallel, TXm and TNm are analysed with Gaussian observation models:

$$Y_{j,t} \mid \mathbf{x}_{j,t}, \Theta_j \sim \mathcal{N}(\mu_{j,t}, \sigma_{j,t}^2). \quad (6)$$

TXx and TNx follow the maximum-GEV distribution

$$G^+(y; \mu, \sigma, \xi) = \exp \left[- \left\{ 1 + \xi \frac{y - \mu}{\sigma} \right\}^{-1/\xi} \right], \quad 1 + \xi \frac{y - \mu}{\sigma} > 0, \quad (7)$$

whereas TXn and TNn follow the corresponding minimum-GEV distribution

$$G^-(y; \mu, \sigma, \xi) = 1 - \exp \left[- \left\{ 1 - \xi \frac{y - \mu}{\sigma} \right\}^{-1/\xi} \right], \quad 1 - \xi \frac{y - \mu}{\sigma} > 0. \quad (8)$$

The continuous Gumbel limits apply at $\xi = 0$. A negative shape parameter implies a finite upper endpoint for a maximum margin and a finite lower endpoint for a minimum margin.

The Gaussian and GEV laws are asymptotically motivated observation models for block means and extrema. Their limits remain available under suitable weak within-block dependence ([Leadbetter et al., 1983](#)). Serial dependence and clustering nevertheless reduce the effective information in a finite block and may slow convergence to those limits, so their adequacy is assessed empirically.

3.2.1 Seasonal dispersion

In parallel with the fixed seasonal cycle in [equation \(5\)](#), we decompose the log observation scale into an overall level and a zero-sum seasonal effect:

$$\log \sigma_{j,t} = \kappa_j + d_{j,s(t)}, \quad \sum_{s=1}^p d_{j,s} = 0. \quad (9)$$

We enforce the zero-sum constraint through

$$\mathbf{d}_j = C_p \mathbf{u}_j, \quad C_p^\top C_p = I_{p-1}, \quad C_p^\top \mathbf{1}_p = \mathbf{0}, \quad (10)$$

where $\mathbf{d}_j = (d_{j,1}, \dots, d_{j,p})^\top$, C_p is a fixed orthonormal contrast matrix, and $\mathbf{u}_j \in \mathbb{R}^{p-1}$ contains the free coefficients. For $p = 4$, three coefficients therefore specify the four seasonal log-scale effects. Thus $\exp(\kappa_j)$ is the geometric mean of the four seasonal scales. Unlike the evolving seasonal location, the $d_{j,s}$ are fixed observation parameters, so the scale pattern repeats across years. For a Gaussian margin, $\sigma_{j,t}$ is the conditional standard deviation of a seasonal mean; for a GEV margin, it is a scale parameter. Neither is the standard deviation of daily temperature.

We likewise keep the GEV shape parameter constant and concentrate temporal evolution in location, our principal scientific target. The exploratory analysis clearly supports seasonal differences in dispersion but not a particular secular scale evolution. Time variation in GEV shape is still harder to identify because shape is learned from limited tail information ([Coles, 2001](#), Chap. 6); constant shape is consequently a common restriction in nonstationary climate-extreme models ([Kharin and Zwiers, 2005](#)). We prespecify this parsimonious model and assess these restrictions using period-stratified diagnostics and sensitivity analyses in [appendix S4](#), as well as an outlook for future model selection directions in the Discussion.

3.3 Priors

We distinguish uncertainty about the starting state, subsequent climatic evolution and variability around the latent location. A prior is judged by what it implies for these quantities in temperature units. Increasing every prior scale would not provide a neutral default: state innovations accumulate through time, and a broad prior on a small per-season coefficient can imply large changes over decades. We therefore calibrate the evolution priors and examine complete prior simulations ([Gelman et al., 2017](#); [Gabry et al., 2019](#)).

For the initial level and the three free coordinates of the initial seasonal cycle, we retain

$$\alpha_{j,0} \sim \mathcal{N}(0, 20^2), \quad \boldsymbol{\gamma}_j^{(0)} \sim \mathcal{N}(\mathbf{0}, 20^2 I_{p-1}). \quad (11)$$

These are broad starting-state priors, separate from the priors regulating subsequent evolution. Their large scale does not establish that the induced seasonal cycles are physically plausible; this is assessed in [appendix S1.5](#). For the observation distributions, we use

$$\exp(2\kappa_j) \sim IG(2, 2), \quad \mathbf{u}_j \sim \mathcal{N}(\mathbf{0}, 0.30^2 I_{p-1}), \quad \xi_j \sim \mathcal{N}(0, 0.30^2). \quad (12)$$

Here $\exp(\kappa_j)$ is the geometric mean of the seasonal observation scales. The vector $\mathbf{u}_j$ contains their orthonormal log-scale contrast coordinates: $\mathbf{d}_j = C_p \mathbf{u}_j$, as defined in [equation \(10\)](#). The shape prior applies only to GEV margins and is not truncated. The complete specification is given in [appendix S1](#).

3.3.1 Calibrated hierarchical shrinkage

In the non-centred representation, the signed innovation coefficients control departures from a deterministic trend and seasonal cycle. Following [Frühwirth-Schnatter and Wagner \(2010\)](#), we

regularize them towards zero with normal priors, while estimating their component-specific scales jointly across the six summaries:

$$\begin{aligned} s_{c,j} \mid \tau_c &\sim \mathcal{N}(0, \tau_c^2), & \tau_c &\sim \text{HalfNormal}(A_c), & c \in \{\alpha, \beta, \gamma\}, \\ \beta_{j,0} &\sim \mathcal{N}(0, P_{\beta_0}). \end{aligned} \quad (13)$$

The half-normal distribution is that of $A_c|Z|$, where $Z \sim \mathcal{N}(0, 1)$. Thus A_c is a fixed calibration scale, τ_c is a learned prior SD, and $|s_{c,j}|$ is the innovation SD of response j . The three hyperpriors are independent. Initial slopes retain separate normal priors with fixed variance P_{β_0} .

Sharing τ_c lets related summaries inform the typical magnitude of each kind of evolution while retaining different process variances and trajectories. It imposes no common warming direction. This is useful when similar location paths can be obtained by different allocations of level and slope variation, but it cannot by itself resolve that ambiguity or ensure prior robustness. Hierarchical scale priors still require calibration and sensitivity assessment (Gelman, 2006; Belmonte et al., 2014; Bitto and Fröhlich-Schnatter, 2019).

Our reference calibration is

$$(A_\alpha, A_\beta, A_\gamma) = (10^{-2}, 10^{-4}, 10^{-2}), \quad P_{\beta_0} = 10^{-4}. \quad (14)$$

Since $\mathbb{E}(\tau_c^2) = A_c^2$, the marginal variance of $s_{c,j}$ is A_c^2 . Over 120 seasonal transitions, the level, integrated-slope and same-season innovation contributions have prior SDs of approximately 0.110 °C, 0.075 °C and 0.077 °C. The initial-rate prior has SD $40\sqrt{P_{\beta_0}} = 0.40$ °C per decade and alone contributes a 30-year displacement SD of 1.20 °C. These scales express a preference for modest changes in evolution, while allowing uncertainty about a persistent initial rate. They are neither estimated changes nor upper bounds on warming.

We use “weakly informative” to describe the aim of this scientific calibration, not to claim that the prior has negligible influence on every latent component. Joint prior simulations include the shared scales, response-specific coefficients, initial states and observation parameters. They expose both the implied temperature evolution and the consequences of the broader starting-state priors. The calculations, simulations and their limitations are reported in [appendix S1](#); [appendix S2](#) compares scale choices, hyperprior tails and unpooled formulations.

3.4 Posterior computation

The conditional structure suggests a blocked Metropolis-within-Gibbs sampler alternating between response-specific trajectories and parameters, followed by the shared shrinkage scales (Fearnhead and Sherlock, 2026). One sweep of the joint posterior in [equation \(1\)](#) is shown in [figure 3](#). Given the current shared scales, the six response-specific blocks can be updated in parallel; the scales are then updated using all six coefficient sets.

For a Gaussian response, the observation equation is linear in the standardized states when the structural coefficients are held fixed. The complete path can therefore be sampled exactly by forward filtering and backward sampling (FFBS) (Carter and Kohn, 1994; Fröhlich-Schnatter, 1994; Durbin and Koopman, 2000, 2012). For a GEV margin, a local Laplace approximation $\tilde{L}$ to the exact conditional likelihood L defines the independence proposal

$$q(\tilde{\mathbf{x}}_j) \propto p_0(\tilde{\mathbf{x}}_j) \tilde{L}(\tilde{\mathbf{x}}_j), \quad (15)$$

which is sampled by FFBS. A proposed complete path $\tilde{\mathbf{x}}_j^*$ is accepted with probability

$$\rho_{\text{MH}}(\tilde{\mathbf{x}}_j, \tilde{\mathbf{x}}_j^*) = \min \left\{ 1, \frac{L(\tilde{\mathbf{x}}_j^*) \tilde{L}(\tilde{\mathbf{x}}_j)}{L(\tilde{\mathbf{x}}_j) \tilde{L}(\tilde{\mathbf{x}}_j^*)} \right\}. \quad (16)$$

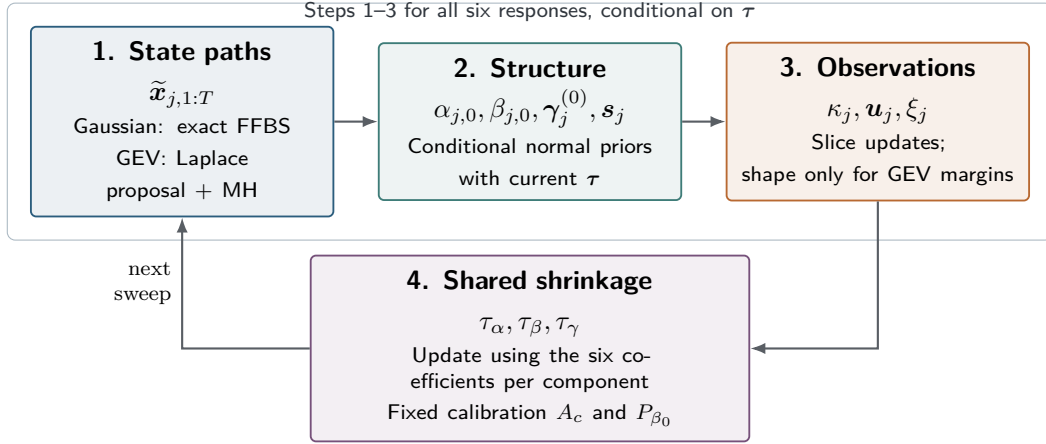


Figure 3: One sweep of the hierarchical posterior sampler. The response-specific path, structural and observation updates are conditional on the current shared scales. A fourth block updates the three shrinkage scales from all six responses. The half-normal calibration scales and initial-slope prior variance stay fixed. Gaussian paths use exact FFBS; GEV paths use a Laplace proposal with an exact MH correction. Independent MCMC chains target the complete hierarchy.

The approximation is held fixed during this update. Because L includes the exact GEV density and support, the correction preserves the specified conditional posterior; approximation quality affects efficiency, not the retained target.

Conditional on the paths, Gaussian structural coefficients have direct Gaussian updates, whereas GEV coefficient blocks use elliptical slice sampling with an exact conditional target (Murray et al., 2010). The remaining observation parameters are updated by elliptical or componentwise stepping-out slice sampling as appropriate (Neal, 2003). The shared shrinkage scales have one-dimensional full conditionals and are updated on the log scale. The complete target, Gaussian state update, proposal construction and shrinkage update are given in [appendix S3](#).

4 Model assessment and predictive performance

We first assess marginal fit using probability integral transform (PIT) histograms and normal-score Q–Q plots. For the conditional CDF $G_{j,t}$ on the original temperature scale, define

$$U_{j,t} = G_{j,t}(Y_{j,t} \mid \mathbf{x}_{j,t}, \Theta_j), \quad Z_{j,t} = \Phi^{-1}(U_{j,t}), \quad (17)$$

where Φ is the standard-normal CDF. Correctly specified continuous conditional margins give the reference distributions

$$U_{j,t} \sim \text{Unif}(0, 1), \quad Z_{j,t} \sim \mathcal{N}(0, 1). \quad (18)$$

PITs computed from posterior-smoothed states reuse the fitted observations and are descriptive residual diagnostics rather than calibrated forecasts. Posterior predictive checks compare observed normal-score dispersion, skewness and tail frequencies with those of replicated scores under the same posterior draws (Gelman et al., 1996). Details are given in [appendix S4](#).

Forecasting follows directly from the state equations. Conditional on a posterior draw of the terminal state and parameters, the mean forecast of the latent location h seasons ahead is

$$\mu_{j,T+h|T} = \alpha_{j,T} + h\beta_{j,T} + \gamma_{j,T+h|T}. \quad (19)$$

Here $\gamma_{j,T+h|T}$ is the seasonal component forecast from the state at T , obtained by propagating the dummy-seasonal transition with future innovations set to zero. The implied cycle repeats every four seasons. Full predictive simulations instead draw future innovations and then Gaussian or

GEV observations, propagating uncertainty in the terminal state, parameters and future evolution together with observation variability. [Appendix S5](#) gives the forecast calculations.

We assess predictive performance through expanding-window refits, using only observations of all six summaries available at each forecast origin; the shared scales are re-estimated within each training window. Forecast calibration is examined using PIT plots and the coverage and average width of central 90%, 95% and 99% prediction intervals. Predictive accuracy is compared using CRPS, with smaller values indicating better performance ([Diebold et al., 1998](#); [Gneiting and Raftery, 2007](#)). The forecast design, seasonal breakdowns and directional tail misses are reported in [appendix S4](#).

5 Tracking evolving risk

The fitted distributions describe both the rarity of a fixed temperature and the temperature associated with a fixed rarity. For the seasonal CDF $G_{j,t}$, upper- and lower-tail probabilities are

$$p_{j,t}^+(c) = 1 - G_{j,t}(c), \quad p_{j,t}^-(c) = G_{j,t}(c). \quad (20)$$

For example, the upper-tail probability for JJA TXx describes the chance of at least one summer day exceeding c . For a specified season, $\mathcal{R}_{j,t}^\pm(c) = 1/p_{j,t}^\pm(c)$ is its stationary-equivalent return period in years. This describes recurrence if that season’s distribution remained unchanged, rather than the expected waiting time under an evolving climate ([Cooley, 2012](#); [Salas and Obeysekera, 2014](#)). Conversely, the upper- and lower-tail r -year return levels, for $r > 1$, are the quantiles

$$z_{j,t}^+(r) = G_{j,t}^{-1}(1 - 1/r), \quad z_{j,t}^-(r) = G_{j,t}^{-1}(1/r). \quad (21)$$

We calculate these quantities within each posterior draw and report medians and pointwise 95% credible intervals. The simulated future paths from [section 4](#) extend the same calculations to forecasts of return periods and levels; averaging the conditional exceedance probabilities gives the predictive event probability.

Calculations for annual summaries and risks over longer forecast windows are given in [appendix S5](#).

6 Results

Computation: update from the separate fits

Insert the completed-run chain lengths, convergence summaries and computational cost. The screen and paper settings are specified in [appendix S2](#); report the settings of the actual fits used for the numerical results.

The whole-path proposals were highly efficient: GEV paths were accepted in 97.2% of updates overall, ranging from 95.2% for TNn to 98.6% for TXn. This high acceptance rate indicates that the Gaussian state-space proposal closely approximated the exact conditional target. Every local optimization used to construct a proposal converged, and no proposal was rejected because the optimizer failed to respect the GEV support.

The posterior GEV shape medians were -0.190 for TXx, -0.230 for TXn, -0.156 for TNx and -0.116 for TNn, with respective 95% credible intervals $(-0.234, -0.141)$, $(-0.275, -0.182)$, $(-0.199, -0.108)$ and $(-0.165, -0.061)$. All four relevant tails are therefore estimated to be bounded, having finite statistical endpoints. Trace plots, Monte Carlo errors (rank-normalized splits and effective sample sizes) and sensitivity checks are provided in [appendices S2](#) and [S4](#).

Seasonal dispersion is clearly response-specific ([figure 4](#)). DJF has the largest median scale for TXm and TNm (1.45°C and 1.46°C), and TNm falls to 0.68°C in JJA. Among the extrema,

TXx is most dispersed in SON (2.70°C), whereas TXn and TNn are again most dispersed in DJF (2.78°C and 3.05°C). Separating this important recurring pattern from the latent location prevents the state evolution from having to reproduce predictable changes in spread.

Figure to insert: `seasonal_scales.png`

Figure 4: Seasonal observation scales. Points and bars show posterior medians and 95% credible intervals; small horizontal offsets separate the summaries. For TXm and TNm, scale is the conditional standard deviation of a seasonal mean. For the extrema it is the GEV scale parameter, not a standard deviation.

The pooled PIT histograms and normal-score Q–Q plots are broadly consistent with their reference distributions (figure 5). The clearest feature is mild left-skewness in the TNm normal scores. Two-sided Kolmogorov–Smirnov tests likewise provide no evidence against uniformity, with p -values of 0.578 (TXm), 0.107 (TNm), 0.948 (TXx), 0.929 (TXn), 0.619 (TNx) and 0.711 (TNn). Because the PITs are obtained in-sample and retain some serial dependence, these tests are interpreted descriptively. Season-stratified diagnostics in Supplementary S4 reveal several localized departures that are obscured by pooling and examine the remaining serial structure.

Figure to insert: `seasonal_pit_qq.png`

Figure 5: Goodness-of-fit diagnostics. For each response, the dashed line indicates the uniform-density reference. The corresponding Q–Q plot compares normal-score-transformed PITs with standard-normal quantiles. These are in-sample diagnostics; season-stratified versions are provided in Supplementary appendix S4.

Prior sensitivity: screen results pending

The complete settings and figure placeholders are in appendix S2. Insert the main findings on levels, warming rates, seasonal evolution and risk probabilities after checking the separate fits.

Expressed on a ten-year horizon, the posterior median standard deviations generated by new level innovations range from 0.085°C to 0.195°C for the different summaries, those generated by accumulated slope innovations from 0.035°C to 0.048°C , and those generated by same-season innovations from 0.016°C to 0.046°C . These quantities describe the stochastic contribution of the state innovations. Their magnitude permits gradual changes in the latent trajectories without allowing them to follow short-term seasonal weather noise.

6.1 Evolution of levels and warming rates

All six fitted levels rise across the record (figure 6). Daytime trajectories flatten or dip around mid-century before rising more strongly from the 1970s, whereas nighttime trajectories increase more steadily.

Figure to insert: `seasonal_levels.png`

Figure 6: Latent temperature levels, 1892–2026. Curves and bands show posterior medians and pointwise 95% credible intervals for TX summaries (top row) and TN summaries (bottom row), in the original temperature orientation.

The slope trajectories describe the local rate of change (figure 7). Their posterior bands show uncertainty in when and how quickly the fitted levels change, particularly near the ends of the record. Sensitivity to the structural priors is examined in appendix S2.

Figure to insert: `seasonal_slopes.png`

Figure 7: Local warming rates. Curves and bands show posterior medians and pointwise 95% credible intervals in degrees Celsius per decade, in the original temperature orientation. Dashed lines mark zero.

6.2 Evolving risk and prediction

The location shifts translate nonlinearly into event probabilities (figure 8). For JJA, the posterior median probability that TXx exceeds 35 °C rises from 3.6% (95% interval 1.7–6.7%) in the first 30 years of the record to 15.9% (11.1–22.0%) in the most recent 30 years and 22.4% (14.2–34.0%) in 2026 . Under conditions held fixed at each of those periods, that would correspond to roughly one summer in 28, one in six and one in four/five.

Figure to insert: `seasonal_risks.png`

Figure 8: Changing seasonal risks. Posterior median event probabilities and pointwise 95% credible intervals for JJA TXx > 35 °C, JJA TNx > 25 °C, DJF TXn < 0 °C and DJF TNn < –10 °C.

The chance of a summer containing a daily minimum above 25 °C also rises, although such a hot night remains unusual: the fitted JJA probability increases from 0.0024% (0–0.072%) in the early period to 0.186% (0.024–0.742%) recently and 0.377% (0.061–1.331%) in 2026. At the cold end, a winter containing an *ice day*—a day whose maximum stays below 0 °C—becomes somewhat less likely: its probability falls from 94.7% (90.9–97.2%) to 79.5% (72.8–85.3%). The probability of a winter minimum below –10 °C falls from 45.8% (38.1–54.5%) to 25.8% (19.5–32.6%). For DJF 2026, the corresponding fitted probabilities are 73.3% and 21.3%.

Uccle’s reported heat record of 36.6 °C, set in 1947, stood for 72 years. By spring 2019, a new record was plausible. Fitted using observations only through May 2019, the model assigned the coming summer an 11.5% probability of exceeding 35 °C (6.5–18.2%) and a 3.1% probability of exceeding the old record (1.1–6.3% [figure 9](#)).

The summer that followed was exceptionally hot. On 25 July, Uccle reached 39.7 °C, surpassing the 1947 benchmark by 3.1 °C. Before the season, the fitted probability of an extreme this high was only 0.046%—about 70 times smaller than the probability of breaking the old record. The posterior median conditional probability was 0.016% (95% interval 0–0.265%). This was a record-shattering extreme ([Fischer et al., 2025](#)).

The heat extended beyond that single daytime maximum: all six observed JJA 2019 summaries exceeded their predictive medians, while TXx and TNx also exceeded the upper limits of their 95% prediction intervals.

Figure to insert: `pre2019_record_event.png`

Figure 9: The 2019 record summer viewed from May 2019. *A:* One-season-ahead predictive medians and 95% prediction intervals for the six JJA summaries; diamonds show the observations. *B:* Posterior mean probability of JJA TXx exceeding each threshold, with pointwise 95% intervals for the conditional probability. The gold point marks 35 °C; the red point and vertical line mark the observed 39.7 °C.

Ten-year forecasts propagate the terminal posterior state through the same structural evolution used within the observed period. Their central tendency continues the latest estimated local trend, while future state innovations allow individual trajectories to depart from that extrapolation. These are statistical forecasts rather than climate projections and are not conditioned on a particular emissions scenario. For JJA 2036, the posterior median probability that TXx exceeds 35 °C is 26.9% (14.9% – 46.5%), while the probability that TNx exceeds 25 °C is 0.58% (0.09% – 2.49%). For DJF 2036, the probabilities that TXn falls below 0 °C and TNn below –10 °C are 68.8% (54.4% – 80.6%) and 18.9% (11.7% – 27.5%), respectively. These intervals propagate uncertainty in the terminal state and model parameters together with future state innovations.

It is important to distinguish uncertainty about the latent climate state from the variability of a future observation. In [figure 10](#), the 95% prediction intervals for future JJA TXm and TXx observations are substantially wider than the corresponding posterior intervals for their latent locations. For TXx, the GEV location parameter is moreover not the predictive median: $G^+(\mu) = \exp(-1)$, so μ represents approximately the 37th percentile of a (maximum) GEV distribution.

The gradual widening of the forecast bands follows from the state transition. Isolating the level and slope innovations, their contribution at a horizon of h seasonal blocks is

$$\text{Var}(\alpha_{j,T+h} \mid \alpha_{j,T}, \beta_{j,T}, \Theta_j) = h s_{\alpha,j}^2 + s_{\beta,j}^2 \frac{h(h-1)(2h-1)}{6}. \quad (22)$$

Although this uncertainty accumulates with the forecast horizon, the estimated innovation scales are small. Over ten years, its contribution is therefore modest relative to the much larger interannual observation variability, whose season-specific scale does not itself increase with the horizon.

The present ten-year display does not yet show the accumulation of state uncertainty sufficiently clearly, because its contribution remains small relative to ordinary weather

variability. We will therefore replace it with a 30-year forecast display, directly addressing the reviewer’s question of how predictive uncertainty broadens with the forecast horizon.

Figure to insert: `forecast_uncertainty_seasonal.png`

Figure 10: Ten-year JJA temperature forecasts. Top: 95% prediction intervals for future TXm and TXx observations (blue) and 95% posterior intervals for their latent locations (red), 2027–2036. Bottom: widths of the corresponding intervals.

7 Discussion

Structural time-series models give an extreme-value distribution an interpretable changing history. The latent level, slope and seasonal cycle describe persistent evolution, while the observation distribution accounts for variability among seasonal summaries. In the Uccle application, cumulative warming is clearer than the detailed history of changes in its rate, particularly near the record boundaries. This distinction matters: a smooth fitted trajectory need not imply a precisely estimated local rate, and its bends do not identify physical causes.

The methodological challenge is learning how strongly the latent components should evolve. Small innovations can have substantial cumulative effects while remaining weakly informed by individual observations. The non-centred representation makes their amplitudes accessible to continuous shrinkage. The sensitivity of the preliminary single-series fits motivated a further step: pooling the strength of regularization across related summaries. A long record can constrain the overall location while leaving substantial ambiguity about how level and slope innovations produce it. Each fitted rate can then depend strongly on the separately chosen prior scales.

The hierarchy lets all six summaries inform three component-specific shrinkage scales and may stabilize the allocation of variation while preserving distinct histories. Whether it achieves this must be established by the sensitivity comparisons; partial pooling is not a guarantee of identification. Its scientific value is reduced sensitivity of the posterior trajectories and risk estimates, rather than narrower intervals by themselves. The unpooled comparisons in [appendix S2.2](#) and the alternative hyperpriors in [appendix S2.3](#) assess this benefit. Only six coefficients inform each shared scale, so borrowing information complements physical calibration and sensitivity assessment; it does not remove their need. Stable level estimates may still coexist with uncertainty about local rates or seasonal evolution.

Calibration gives these priors a scientific interpretation. The same numerical scale can imply very different temperature changes for a level random walk, an integrated slope or a seasonal component. Weak informativeness therefore concerns their implications within the model, not the number of decimal places in a coefficient or an arbitrarily large prior variance ([Gelman et al., 2017](#); [Gabry et al., 2019](#)). The reference favours modest departures from a persistent trend and seasonal cycle; the sensitivity analysis tests whether the resulting trajectories and risks depend materially on that preference. Hyperpriors extend this calibration by allowing the responses to learn common evolution scales, while their tail behaviour remains an explicit modelling choice ([Gelman, 2006](#)).

The joint prior simulations also reveal a limitation of the retained starting-state specification. The SD-20 seasonal-coordinate priors generate very broad annual cycles and substantial mass on

implausible temperature patterns. This finding is separate from the much smaller prior changes in seasonality over 30 years. It motivates assessing a tighter, physically interpretable initial-cycle prior, including its coordinate dependence, rather than broadening stochastic evolution scales to address every source of sensitivity. The current simulations provide a diagnostic of the stated model; they do not certify all of its priors as weakly informative or make its six replicated summaries physically ordered.

The state update exploits the latent Gaussian structure while retaining the GEV observation law. Gaussian simulation smoothing supplies a complete-path proposal, and the Metropolis–Hastings correction preserves the posterior target. High acceptance in the present application supports the usefulness of this proposal; it does not replace convergence checks or establish its efficiency for every extreme-value likelihood. Comparisons with particle MCMC would help determine how performance changes with stronger nonlinearities, longer records or observations close to finite endpoints (Andrieu et al., 2010; Lindsten et al., 2014).

The risk calculations connect distributional evolution to events with a direct temperature interpretation. A familiar threshold can become much more likely while an extreme overshoot remains exceptional, as illustrated by the 2019 summer (Fischer et al., 2025). Return periods must therefore be read as stationary equivalents of the fitted seasonal probabilities. Forecasts propagate uncertainty in the current state and subsequent evolution, but extrapolate statistical dynamics rather than an emissions scenario.

The summaries are derived from the same days and are likely to retain positive residual correlations, especially between daytime and nighttime temperatures. Sharing prior scales does not represent this residual association. Our product of conditional observation likelihoods is therefore a working assumption whose effect on the uncertainty of pooled estimates requires assessment. A richer observation model would also be needed for formal contrasts between summaries, since uncertainty in a difference depends on their covariance. Overlap of marginal credible intervals cannot establish equal warming.

Several modelling restrictions warrant further assessment. Observation scales repeat by season, and the GEV shapes are constant through time. Changes in these features may matter for remote thresholds even when the location describes most historical variation. Seasonal maxima also do not describe the duration or clustering of heatwaves. Prior sensitivity, posterior predictive checks and held-out forecasts address different parts of model adequacy; none alone establishes reliable extrapolation far beyond the observed range.

Continuous shrinkage also leaves open which components should evolve at all (Slater et al., 2021). It allows innovation scales arbitrarily close to zero without assigning probability to an exactly fixed component. Formal selection between fixed and evolving levels, slopes or seasonal cycles would require a separate model-selection construction, for example point masses or dimension-changing moves (Frühwirth-Schnatter and Wagner, 2010; Cadonna et al., 2020; Green, 1995). Near the deterministic limit, such procedures must distinguish absent evolution from evolution too weak to detect, and should be judged through their effects on the scientific quantities of interest.

The framework can be adapted to other environmental variables whose distributional features evolve. Hydrological extremes (Katz et al., 2002) and precipitation applications may require changing occurrence, scale or shape alongside location (Asadieh and Krakauer, 2015; Westra et al., 2013; Bhatia and Ganguly, 2019; Gu et al., 2017; Jorgensen and Nielsen-Gammon, 2024; Jayaweera et al., 2025; Razi Ghalavand et al., 2025). Snow loads and wave climates provide further applications (Le Roux et al., 2022; Wang et al., 2004), while drought requires particular attention to persistence and accumulation (Van Loon, 2015). An r -largest analysis could use more than one extreme per block; threshold and runs models could address event clustering and duration (Coles, 2001). Additional sites or physical covariates could provide regional context (Scott and Varian, 2014). Each extension calls for suitable observation and state components, renewed prior calibration and predictive validation.

8 Conclusion

Bayesian structural models make evolving environmental risks estimable while learning the strength of persistent change. Combining interpretable latent components, calibrated hierarchical shrinkage and an exact correction to Gaussian state proposals provides a practical framework for extremes, with Gaussian models for averages considered alongside them. Related responses inform the regularization while retaining their own trajectories. Analytic calibration and joint prior simulation make the implied magnitudes of evolution explicit and distinguish them from broad uncertainty about the starting state. The Belgian temperature application illustrates how the framework tracks warming and changing threshold probabilities while representing uncertainty in local rates and future conditions. Extensions to other hazards and to selection of the evolving components offer further opportunities for state-space extreme-value analysis.

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Data and code availability

BUCEX, the open-source Python package used for this study, is available at <https://github.com/baaelbre/bucex>. The scripts, configurations and documentation required to reproduce the

analyses are provided under `research/serra`. The six derived seasonal temperature series are distributed with the code.

The original daily record is provided by the Royal Meteorological Institute of Belgium (RMI) and cannot be redistributed under its conditions of use. It may be requested directly from the RMI. Part of the record is freely available online from 1973 onward through Meteociel at https://www.meteociel.fr/climatologie/obs_villes.php?code=6447.

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Supplementary material

S1 State-space representation and prior calibration

S1.1 State-space matrices

For the seasonal model, $p = 4$ and

$$\mathbf{x}_{j,t} = (\alpha_{j,t}, \beta_{j,t}, \gamma_{j,t}, \gamma_{j,t-1}, \gamma_{j,t-2})^\top.$$

The component equations in [equation \(4\)](#) can be written

$$\mathbf{x}_{j,t} = F\mathbf{x}_{j,t-1} + L_j\boldsymbol{\varepsilon}_{j,t}, \quad \mu_{j,t} = H\mathbf{x}_{j,t}, \quad (\text{S1})$$

with

$$F = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad L_j = \begin{pmatrix} s_{\alpha,j} & 0 & 0 \\ 0 & s_{\beta,j} & 0 \\ 0 & 0 & s_{\gamma,j} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad H = (1, 0, 1, 0, 0),$$

and $\boldsymbol{\varepsilon}_{j,t} \sim \mathcal{N}_3(\mathbf{0}, I_3)$. The zero rows in L_j shift the seasonal lags deterministically; the transition covariance of the full companion state is therefore singular, although the three active innovations have an ordinary Gaussian density.

For the non-centred form, define zero-initialized standardized processes

$$\begin{aligned} \tilde{\alpha}_{j,t} &= \tilde{\alpha}_{j,t-1} + \varepsilon_{\alpha,j,t}, \\ \tilde{\beta}_{j,t} &= \tilde{\beta}_{j,t-1} + \varepsilon_{\beta,j,t}, \\ \tilde{A}_{j,t} &= \tilde{A}_{j,t-1} + \tilde{\beta}_{j,t-1}, \\ \tilde{\gamma}_{j,t} &= -\sum_{\ell=1}^{p-1} \tilde{\gamma}_{j,t-\ell} + \varepsilon_{\gamma,j,t}. \end{aligned} \quad (\text{S2})$$

The centred components are recovered as

$$\begin{aligned} \alpha_{j,t} &= \alpha_{j,0} + t\beta_{j,0} + s_{\alpha,j}\tilde{\alpha}_{j,t} + s_{\beta,j}\tilde{A}_{j,t}, \\ \gamma_{j,t} &= \gamma_{j,s(t)}^{(0)} + s_{\gamma,j}\tilde{\gamma}_{j,t}. \end{aligned} \quad (\text{S3})$$

Substitution into $\mu_{j,t} = \alpha_{j,t} + \gamma_{j,t}$ gives [equation \(5\)](#). The signs of $s_{c,j}$ are computational parameters; $|s_{c,j}|$ is the innovation standard deviation and $q_{c,j} = s_{c,j}^2$ is the process variance.

S1.2 Complete prior specification

One transition represents one meteorological season. A decade contains 40 transitions, and the 30-year calibration horizon contains $H = 120$. Each summary has its own coefficients, latent trajectory and observation parameters. The three shrinkage scales $\tau_\alpha, \tau_\beta, \tau_\gamma$ are shared; the initial-slope variance P_{β_0} remains fixed.

[Table S1](#) gives the reference specification. Conditional on the shared scales, the coefficient priors are independent across components and responses. The three hyperpriors, initial-state priors and observation-parameter priors are mutually independent.

The 20 °C standard deviations on the fixed intercept and initial seasonal coordinates express broad uncertainty about the starting state. The seasonal prior is placed on the three lag-state coordinates at the first observation. If these are (g_1, g_2, g_3) , the chronological cycle is

Table S1: Reference priors for the seasonal hierarchy. A_c are fixed hyperprior scales; τ_c are learned shared prior SDs. The initial slope has fixed prior variance P_{β_0} . Shape priors apply only to GEV observations.

Quantity	Reference prior
Initial level $\alpha_{j,0}$	$\mathcal{N}(0, 20^2)$
Initial seasonal coefficients $\gamma_j^{(0)}$	$\mathcal{N}(\mathbf{0}, 20^2 I_{p-1})$
Signed innovation $s_{c,j} \mid \tau_c$	$\mathcal{N}(0, \tau_c^2)$, $c \in \{\alpha, \beta, \gamma\}$
Shared prior SD τ_c	HalfNormal(A_c), independently across components
Initial slope $\beta_{j,0}$	$\mathcal{N}(0, P_{\beta_0})$
Fixed calibration	$(A_\alpha, A_\beta, A_\gamma) = (10^{-2}, 10^{-4}, 10^{-2})$, $P_{\beta_0} = 10^{-4}$
Squared geometric-mean scale $\exp(2\kappa_j)$	$IG(2, 2)$
Seasonal log-scale coordinates $\mathbf{u}_j$	$\mathcal{N}(\mathbf{0}, 0.30^2 I_{p-1})$
GEV shape ξ_j	$\mathcal{N}(0, 0.30^2)$ on $\mathbb{R}$

$(g_1, -g_1 - g_2 - g_3, g_3, g_2)$. The record begins in MAM, so the induced JJA effect has prior SD $20\sqrt{3}^\circ\text{C}$, whereas the other three effects have SD 20°C . This initialization is therefore not an exchangeable prior on four seasonal effects. It also differs from the orthonormal contrast prior on observation log scales. For $V_j = \exp(2\kappa_j)$, $IG(a, b)$ denotes the density proportional to $v^{-a-1} \exp(-b/v)$ for $v > 0$.

Sharing τ_c makes the magnitudes of the response-specific coefficients dependent a priori after integrating over that scale. Their signs remain symmetric, and the standardized state innovations are independent. No common state trajectory or common process variance is imposed. The normal and half-normal priors are continuous, so exactly zero evolution is a limiting model rather than an event with positive posterior probability.

S1.3 Evolution variances in earlier dynamic GEV models

The earlier formulations help distinguish uncertainty about a latent trajectory from uncertainty about its smoothing parameters. Gaetan and Grigoletto (2004) interpreted Gaussian random walks as smoothness priors on changing GEV parameters. Their applications used selected, fixed innovation variances, with posterior predictive assessment and comparisons between alternative values. Fearnhead et al. (2010, Section 5) selected a trend-evolution variance by comparing particle estimates of the model likelihood over a grid. Their subsequent Bayesian calculation incorporated uncertainty in the GEV scale and shape while retaining the selected evolution variance. These analyses therefore differ from integrating over the innovation scales as part of the joint posterior.

Huerta and Sansó (2007) used a different observation hierarchy. In their notation,

$$Y_t \mid \mu_t, \sigma, \xi \sim \text{GEV}(\mu_t, \sigma, \xi), \quad \mu_t = F_t^\top \theta_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, V),$$

with Gaussian state evolution for θ_t and evolution covariance W . Conditional on the intermediate locations μ_t , the structural states can be sampled by Gaussian forward filtering and backward simulation. The authors use inverse-gamma priors for V , and specify W through discount factors or relations to V . This adds short-term variation in the GEV location alongside the state innovations and the GEV observation dispersion. After integrating out ϵ_t , the distribution conditional on θ_t is a Gaussian location mixture of GEV distributions. The distinction matters for interpreting how variability is allocated among the model components. Our location is determined directly by the structural state; the Gaussian approximation used in its update changes the proposal distribution, not the observation hierarchy.

The random-walk model of Wei and Huerta (2017) links the GEV location directly to its preceding value through an innovation with variance V . Their centred MCMC formulation includes an update of V under a normal prior on $\log V$. For their particle-filter comparisons, however, they fix V at a preliminary MCMC posterior mean, reporting instability when estimating

it jointly with the GEV parameters within those filters. The simulation in their Section 4 uses $V = 1$ and GEV scale $\sigma = 1.1$, giving $\sqrt{V}/\sigma \simeq 0.91$. Here σ is a GEV scale, not its observation standard deviation. This example therefore does not by itself assess behaviour when the state-innovation scale is very small relative to the observation scale. Their MCMC comparisons also use 400,000 iterations with 300,000 discarded as burn-in and thinning by 50, and they report difficulties with shorter runs. These findings motivate examining weak evolution explicitly; they do not establish that every centred implementation fails in that regime.

S1.4 Calibration over 30 years

Under $s_{c,j} \mid \tau_c \sim \mathcal{N}(0, \tau_c^2)$ and $\tau_c = A_c |Z_c|$, $Z_c \sim \mathcal{N}(0, 1)$, the marginal coefficient variance is

$$v_c = \mathbb{E}(s_{c,j}^2) = \mathbb{E}(\tau_c^2) = A_c^2, \quad c \in \{\alpha, \beta, \gamma\}. \quad (\text{S4})$$

Thus A_c is the marginal root-mean-square signed coefficient, not the median or standard deviation of τ_c . The initial-slope variance is $\mathbb{E}(\beta_{j,0}^2) = P_{\beta_0}$.

Over H transitions, new level and slope innovations contribute

$$\Delta_{j,H}^{(\alpha)} = s_{\alpha,j} \sum_{r=1}^H \varepsilon_{\alpha,j,r}, \quad \Delta_{j,H}^{(\beta)} = s_{\beta,j} \sum_{r=1}^{H-1} (H-r) \varepsilon_{\beta,j,r}.$$

The standardized innovations are independent of the coefficients. After integrating over both levels of the shrinkage hierarchy,

$$\begin{aligned} \text{sd}_{\pi}(\Delta_{j,H}^{(\alpha)}) &= A_{\alpha} \sqrt{H}, \\ \text{sd}_{\pi}(\Delta_{j,H}^{(\beta)}) &= A_{\beta} \sqrt{\frac{H(H-1)(2H-1)}{6}}. \end{aligned} \quad (\text{S5})$$

For a same-season comparison with H a multiple of $p = 4$,

$$\text{sd}_{\pi}(\Delta_{j,H}^{(\gamma)}) = A_{\gamma} \sqrt{\frac{2H}{p}}. \quad (\text{S6})$$

For the initial rate and its displacement, respectively,

$$\text{sd}_{\pi}(40\beta_{j,0}) = 40\sqrt{P_{\beta_0}}, \quad \text{sd}_{\pi}(H\beta_{j,0}) = H\sqrt{P_{\beta_0}}. \quad (\text{S7})$$

At $H = 120$, the reference calibration gives

$$\begin{aligned} A_{\alpha} = 0.01 &\implies 0.01\sqrt{120} \simeq 0.110^{\circ}\text{C}, \\ A_{\beta} = 0.0001 &\implies 0.0001\sqrt{120 \cdot 119 \cdot 239/6} \simeq 0.075^{\circ}\text{C}, \\ A_{\gamma} = 0.01 &\implies 0.01\sqrt{60} \simeq 0.077^{\circ}\text{C}, \\ P_{\beta_0} = (0.40/40)^2 = 10^{-4} &\implies 0.40^{\circ}\text{C per decade}. \end{aligned} \quad (\text{S8})$$

The initial slope contributes a prior SD of 1.20°C over 30 years. Combining its variance with the three innovation contributions gives a prior SD of 1.21°C for an initial-to-30-year same-season location change. These symmetric scales exclude observation noise. They are standard deviations of non-Gaussian mixtures, not Gaussian 68% intervals. Later windows inherit innovations before their starting date; full prior simulation includes that additional uncertainty.

The conditional SD of an innovation contribution, given τ_c but before drawing its coefficient, is the corresponding gain above multiplied by τ_c . Given $s_{c,j}$, it is instead the gain multiplied by $|s_{c,j}|$. Distinguishing these two quantities avoids confusing a shared prior scale with a fitted process scale.

S1.5 Joint prior simulation and its interpretation

Analytic SDs provide a useful scale check, but they do not describe the full prior distribution of a path or its observations. In particular, Gaussian coefficients mixed over a shared scale produce distributions that are not Gaussian, and the GEV observation law can amplify tail uncertainty. Following [Gabry et al. \(2019, Section 3\)](#), we assess the complete prior-generating model:

$$p(\mathbf{y}^{\text{rep}}) = \int p(\mathbf{y}^{\text{rep}} \mid \mathbf{x}, \Theta) p(\mathbf{x} \mid \Theta) p(\Theta) d\mathbf{x} d\Theta. \quad (\text{S9})$$

No observed temperature or posterior estimate enters this calculation. The historical calendar determines only the number and ordering of blocks. The objective is to understand the magnitude and character of simulated temperature behaviour, rather than to reproduce the observed record closely or select the prior with the highest apparent fit.

For each replication, we draw each of the three shared scales once. We then draw the six response-specific coefficient sets conditionally on those scales, followed by their private initial levels, rates and seasonal states, and their observation parameters. Independent standardized innovations propagate all six state paths. Finally, we sample Gaussian means and GEV extrema conditional on those paths, reflecting lower-extreme responses back to the original temperature scale. A shared scale must not be redrawn independently for each response: that would simulate a different hierarchy.

Each replication covers the 538 observed seasonal positions and a further 120 transitions. The continuation is unconditional prior evolution; it is not a forecast conditional on the temperature record. We separately record the level, integrated-slope and seasonal innovation contributions, initial rate, complete location and replicated observations. Changes at 10, 20 and 30 years from initialization are distinguished from changes over the final 30-year continuation, which also inherit all earlier slope innovations. The same-season calibration removes the repeating initial cycle from a change, but the complete datasets retain that cycle and its uncertainty.

We use 2,000 joint replications for screening and 10,000 for the reported prior figures, with reproducible seeds. Intervals are empirical quantiles: 95% intervals are the main summary, with 99% intervals to inspect tails. They are pointwise prior intervals, not simultaneous path bands or posterior credible intervals. For half-Cauchy scales, finite simulation quantiles remain meaningful even though population variances do not exist. We do not interpret an empirical SD from a finite sample as a finite prior SD.

The reference simulation illustrates why innovation calibration and starting-state calibration must be assessed separately. For TXm, the central 95% intervals of the three 30-year innovation contributions are approximately $[-0.23, 0.23]^\circ\text{C}$, $[-0.16, 0.16]^\circ\text{C}$ and $[-0.17, 0.17]^\circ\text{C}$. Including the initial rate gives an interval of approximately $[-2.34, 2.42]^\circ\text{C}$ for the combined same-season change. These are Monte Carlo summaries of the reference prior, not evidence about the amount of historical warming. The near-zero medians reflect symmetric priors on the direction of change.

The broad starting-state priors generate much more extreme behaviour than is apparent from the innovation SDs alone. The first-year range of the TXm latent location has a simulated median of about 52°C , with a central 95% interval of approximately $[15, 121]^\circ\text{C}$. As a transparent scale audit, we also count temperatures outside $[-50, 60]^\circ\text{C}$. These are illustrative audit thresholds, not likelihood bounds or a claim that every value inside them is plausible for Uccle. Depending on the summary, roughly 8–9% of individual replicated blocks fall outside this range, and 38–49% of the simulated full paths contain at least one such value. These checks identify the SD-20 initialization as deliberately broad; they do not support describing the complete reference prior as physically well calibrated. Under the existing initial-seasonal SD of 2.25°C used as a sensitivity setting, the corresponding range has median 5.9°C and 95% interval $[1.6, 13.6]^\circ\text{C}$. This comparison isolates the initial-cycle scale; it does not establish which setting is scientifically appropriate. The assessment in [appendix S2](#) treats this issue separately from stochastic seasonal evolution.

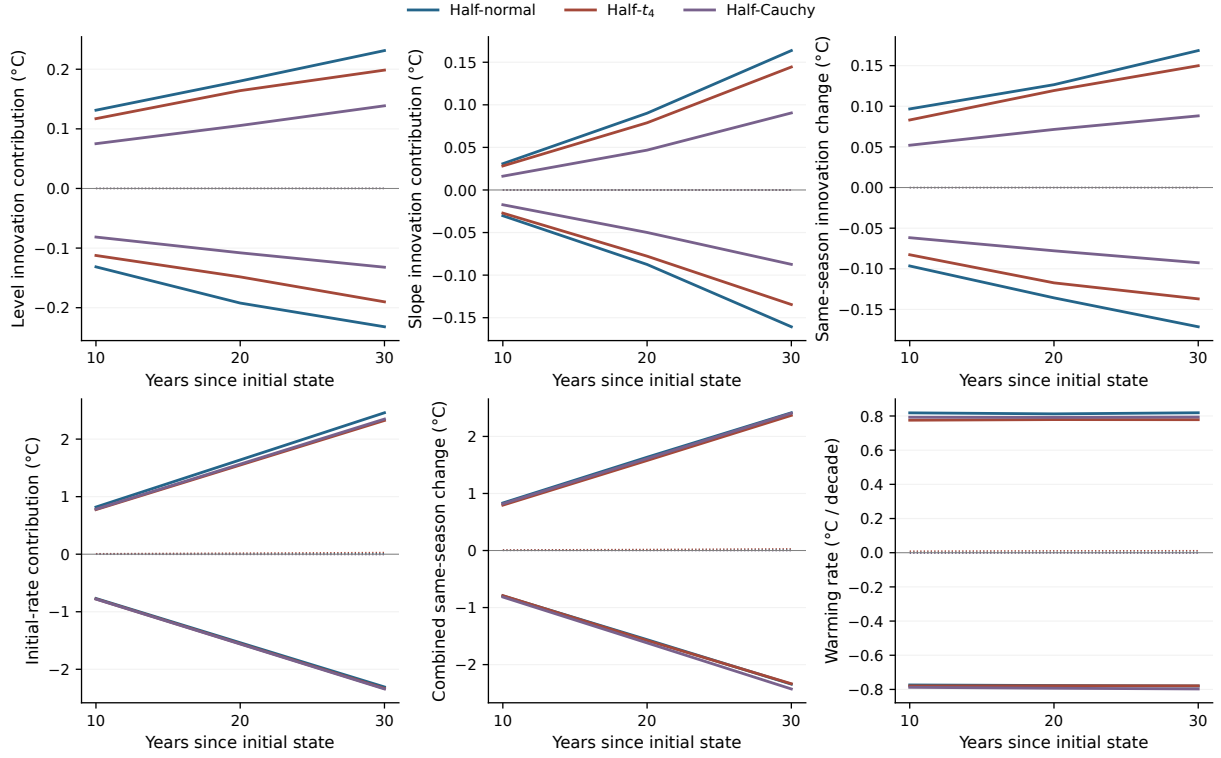


Figure S1: Prior evolution under the reference half-normal and the calibrated half- t_4 and half-Cauchy alternatives. Solid curves connect pointwise 2.5th and 97.5th percentiles at 10, 20 and 30 years; dotted curves connect medians. Initial rates have the same prior in all three families; small differences in their simulated intervals are Monte Carlo variation. The response-specific structural priors are identical, so one response (TXm) is displayed. The bottom-right panel shows the complete rate, including its initial value. All panels use 10,000 joint prior replications; no posterior fit is involved.

We additionally report within-block ordering violations, for example $\text{TXn} \leq \text{TXm} \leq \text{TXx}$. The working product observation model does not enforce such inequalities, and the broad, separately centred initial priors produce frequent violations. We retain these draws: sorting them or discarding incompatible replications would change the specified prior-generating model. This diagnostic concerns the joint physical compatibility of the six summaries and cannot be repaired merely by estimating shared shrinkage scales. Nonfinite simulated values, if encountered, are counted explicitly rather than silently removed from quantile summaries.

S1.6 Observation priors

Write the fixed seasonal log-scale effects as $\mathbf{d}_j = C_p \mathbf{u}_j$, where $C_p^\top C_p = I_{p-1}$ and $C_p^\top \mathbf{1}_p = \mathbf{0}$. Under $\mathbf{u}_j \sim \mathcal{N}(\mathbf{0}, \tau_d^2 I_{p-1})$,

$$\text{Cov}(\mathbf{d}_j) = \tau_d^2 \left(I_p - \frac{\mathbf{1}_p \mathbf{1}_p^\top}{p} \right). \quad (\text{S10})$$

With $p = 4$ and $\tau_d = 0.30$, each $d_{j,s}$ has prior standard deviation $0.30\sqrt{3/4} = 0.260$. Conditional on $\exp(\kappa_j)$, the approximate central 95% prior range for a single seasonal scale is 0.60 to 1.66 times the geometric mean.

The $\mathcal{N}(0, 0.30^2)$ shape prior permits both bounded and heavy-tailed GEV distributions; the likelihood continues to enforce the support condition for every observation. Because the prior does not guarantee finite GEV moments, tail inference is summarized primarily through quantiles and event probabilities.

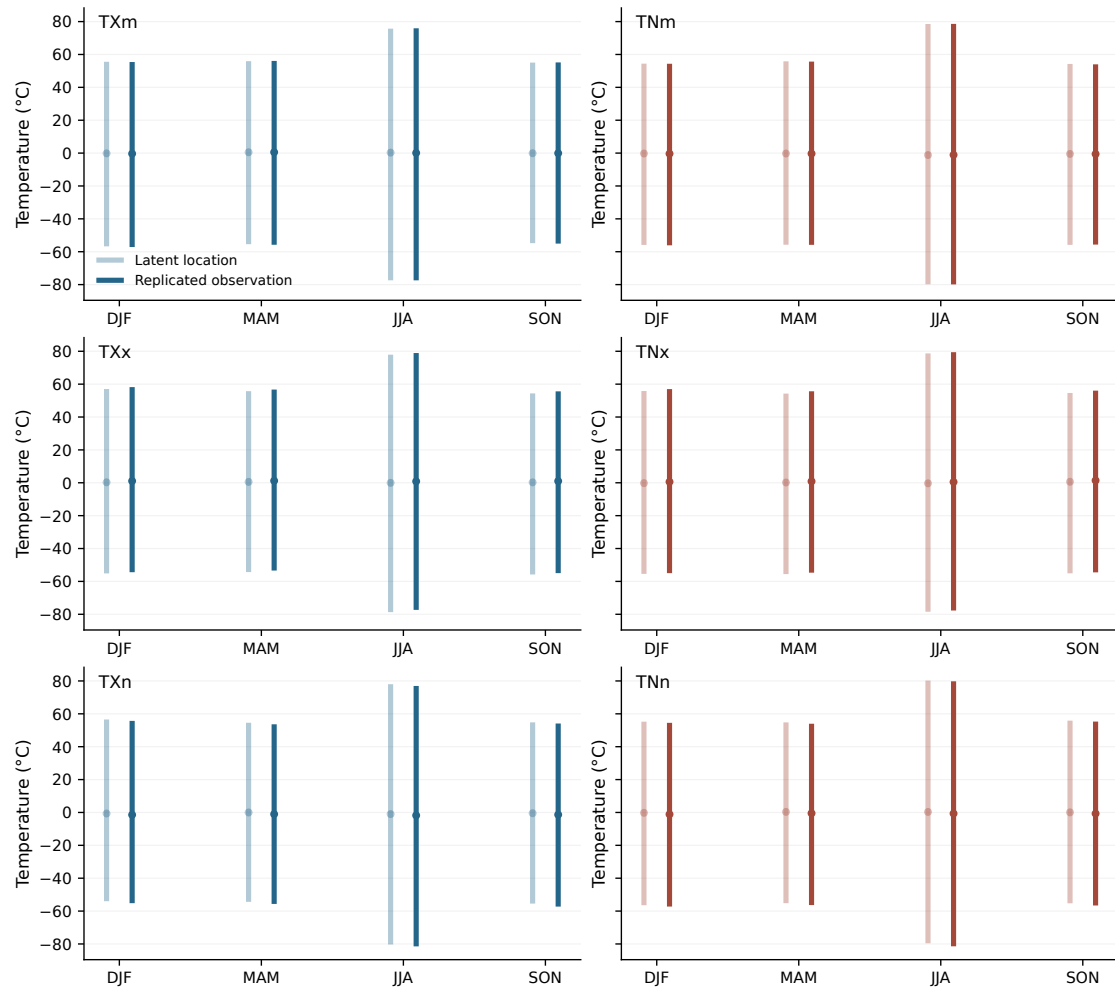


Figure S2: Reference prior intervals for the first annual cycle. Points show medians and vertical segments show central 95% intervals from 10,000 joint replications. Pale and dark segments refer to latent locations and replicated observations, respectively. The larger JJA spread follows from the lag-coordinate initialization described in the text. The plot assesses the full starting-state and observation priors; it does not display fitted seasonal temperatures.

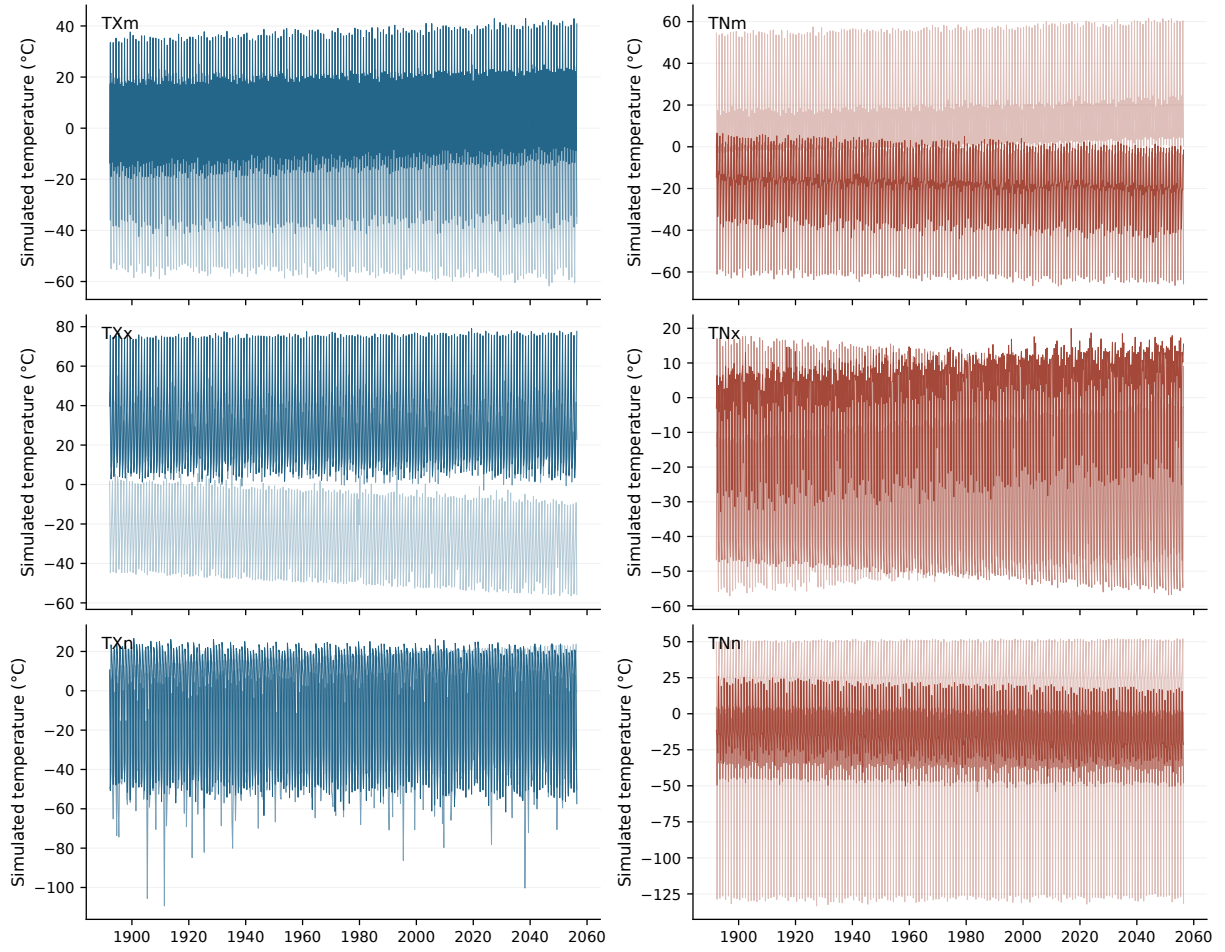


Figure S3: Three complete reference prior replications, shown in their generated order without selection for apparent plausibility. Each includes the historical calendar and a further 30 years. All initial-state and observation uncertainty is propagated. The vertical scales vary between panels to retain the simulated values. These are unconditional synthetic datasets, not reconstructions or forecasts.

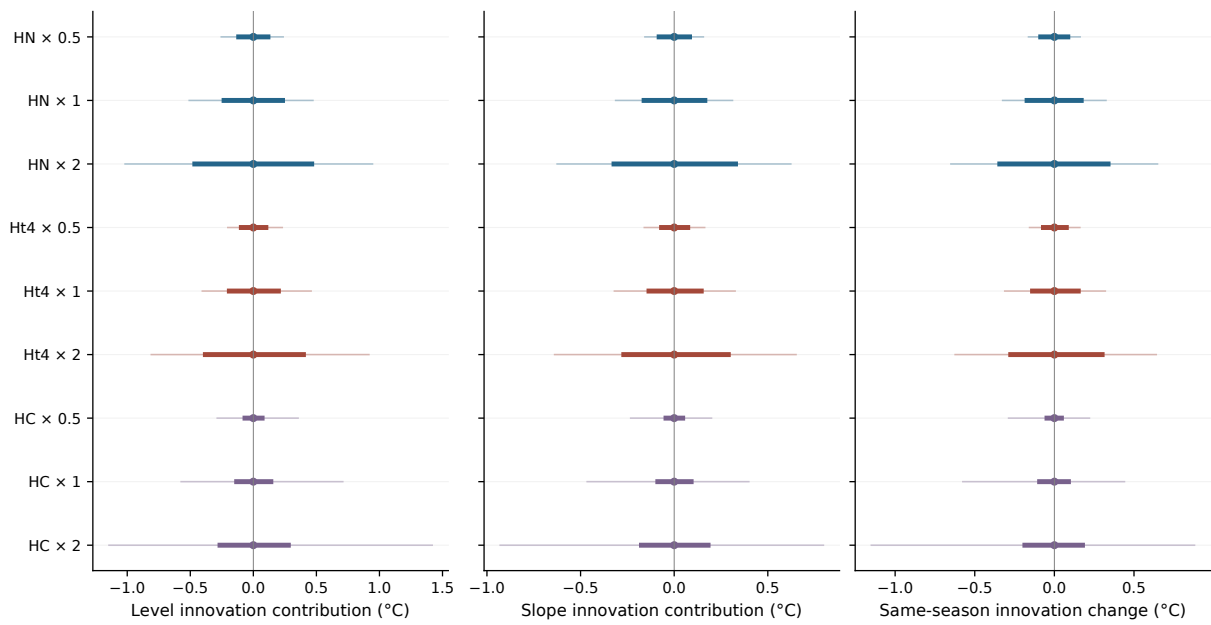


Figure S4: Thirty-year innovation effects under the three hyperprior families and scale multipliers of one half, one and two. HN, Ht4 and HC denote half-normal, half- t_4 and half-Cauchy scales. Thick and thin intervals contain 95% and 99% of prior draws, respectively; points mark medians. Initial-rate priors are unchanged. Half- t_4 matches the reference second moments, whereas half-Cauchy matches the 95th percentile of the shared scale; these are different calibration criteria.

S2 Sensitivity analysis

S2.1 Reference specification and perturbations

The reference R is the pooled half-normal hierarchy in [equation \(13\)](#). The fixed calibration is $(A_\alpha, A_\beta, A_\gamma) = (0.01, 0.0001, 0.01)$, with $P_{\beta_0} = 0.0001$. Each τ_c is estimated from all six responses, while each $s_{c,j}$ and latent path remains response-specific. Sensitivity separates changes to the physical calibration, removal of pooling, and changes to the hyperprior family.

[Table S2](#) retains the reference and 22 calibration or structural alternatives. Half/double factors act on the three hyperprior scales and on $\sqrt{P_{\beta_0}}$, not on fixed process variances. Halving or doubling the initial-slope prior SD divides or multiplies its variance by four, giving initial-rate prior SDs of 0.20°C , 0.40°C and 0.80°C per decade. Additional seasonal and slope combinations examine how variation is allocated across components. Fixed seasonality retains an estimated repeating cycle but removes its stochastic evolution and the inactive τ_γ from that fit.

Table S2: Calibration and structural sensitivity settings for the pooled reference. The first three columns are half-normal hyperprior scales; the fourth is the fixed initial-slope prior SD. All unlisted settings equal the reference. In the final column, τ_d is the prior SD of the seasonal log-scale coordinates and $\sigma_{\gamma,0}$ is the prior SD of the three free initial seasonal coordinates. The reference has $\tau_d = 0.30$, $\sigma_{\gamma,0} = 20$, evolving location seasonality, season-specific observation scales and an unbounded $\xi \sim \mathcal{N}(0, 0.30^2)$ prior. A dash denotes an inactive component.

Setting	A_α	A_β	A_γ	$\sqrt{P_{\beta_0}}$	Other change
R: Reference	0.01	0.0001	0.01	0.01	None
S1: All four scales halved	0.005	0.00005	0.005	0.005	None
S2: All four scales doubled	0.02	0.0002	0.02	0.02	None
S3: Level anchor halved	0.005	0.0001	0.01	0.01	None
S4: Level anchor doubled	0.02	0.0001	0.01	0.01	None
S5: Slope anchor halved	0.01	0.00005	0.01	0.01	None
S6: Slope anchor doubled	0.01	0.0002	0.01	0.01	None
S7: Seasonal anchor halved	0.01	0.0001	0.005	0.01	None
S8: Seasonal anchor doubled	0.01	0.0001	0.02	0.01	None
S9: Initial-slope SD halved	0.01	0.0001	0.01	0.005	None
S10: Initial-slope SD doubled	0.01	0.0001	0.01	0.02	None
S11: Seasonal anchor quartered	0.01	0.0001	0.0025	0.01	None
S12: Double slope, half seasonal anchor	0.01	0.0002	0.005	0.01	None
S13: Double slope, quarter seasonal anchor	0.01	0.0002	0.0025	0.01	None

Continued on the next page

Table S2 (continued)

Setting	A_α	A_β	A_γ	$\sqrt{P_{\beta_0}}$	Other change
S14: Narrow shape prior	0.01	0.0001	0.01	0.01	$\xi \sim \mathcal{N}(0, 0.15^2)$, unbounded
S15: Wide shape prior	0.01	0.0001	0.01	0.01	$\xi \sim \mathcal{N}(0, 0.60^2)$, unbounded
S16: Bounded normal shape prior	0.01	0.0001	0.01	0.01	$\mathcal{N}(0, 0.30^2)$ truncated to $[-0.5, 0.5]$
S17: Uniform shape prior	0.01	0.0001	0.01	0.01	$\xi \sim \text{Unif}(-0.5, 0.5)$
S18: Constant dispersion	0.01	0.0001	0.01	0.01	One observation scale across seasons
S19: Narrow log-scale prior	0.01	0.0001	0.01	0.01	$\tau_d = 0.15$
S20: Wide log-scale prior	0.01	0.0001	0.01	0.01	$\tau_d = 0.60$
S21: Fixed location seasonality	0.01	0.0001	—	0.01	Repeating cycle; no seasonal innovations
S22: Tighter initial seasonal prior	0.01	0.0001	0.01	0.01	$\sigma_{\gamma,0} = 2.25$

The shape-prior changes apply directly only to the four GEV responses. They can nevertheless affect the two Gaussian means indirectly through the shared shrinkage scales, so those means are not unchanged controls under S14–S17. All six responses must be included in each pooled refit. The observation-scale and initial-level priors otherwise stay as in [table S1](#).

S2.2 What happens without pooling?

The preliminary separate-series sensitivity motivated this comparison: similar fitted locations can coexist with different rate histories when the prior reallocates change between level and slope innovations. The purpose is to determine whether sharing the regularization reduces this sensitivity, and which quantities remain weakly identified.

The fixed-prior comparison U uses six separate analyses with

$$s_{c,j} \sim \mathcal{N}(0, A_c^2), \quad \beta_{j,0} \sim \mathcal{N}(0, P_{\beta_0}),$$

and no learned shared scale. Its coefficient variances and 30-year effect SDs match those of R, but its marginal coefficient distributions differ. We retain the full S1–S22 perturbation grid for these separate fits, denoting the resulting comparisons U1–U22. Each uses the same numerical anchors as the corresponding row, now as fixed normal prior SDs.

A second comparison L isolates information sharing more closely:

$$s_{c,j} \mid \tau_{c,j} \sim \mathcal{N}(0, \tau_{c,j}^2), \quad \tau_{c,j} \sim \text{HalfNormal}(A_c)$$

independently for every response. This has the same marginal coefficient prior as R but removes the shared hyperparameters. Each local scale is informed by only one innovation coefficient, so uncertainty about it is expected; it is included as a control for pooling rather than as a more parsimonious model. The comparisons R versus U and R versus L answer different questions and are reported separately.

We examine whether level trajectories are stable while rate trajectories, $|s_{\alpha,j}|$, $|s_{\beta,j}|$ or their posterior association change across calibrations. Such behaviour indicates uncertainty in the

decomposition of persistent change, not necessarily a failure to estimate the total location. Seasonal evolution is displayed separately, so that stability of rates cannot conceal excessive shrinkage of the seasonal component. All comparisons also retain the threshold risks and forecasts.

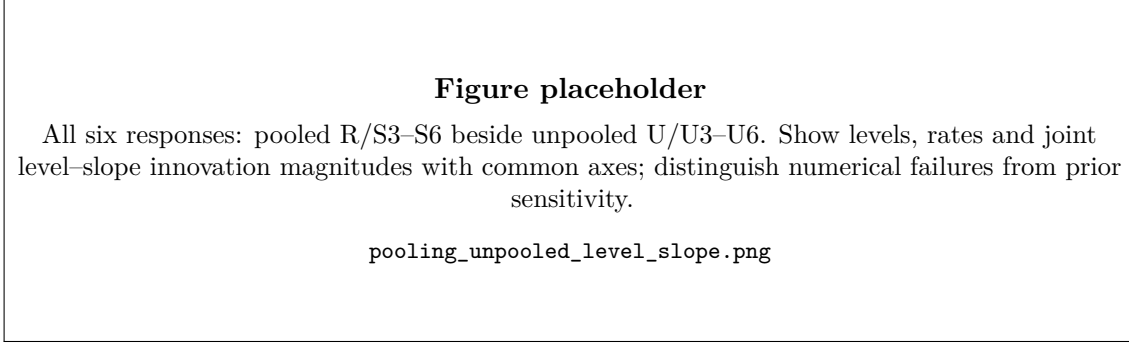


Figure S5: Level–slope ambiguity with and without shared shrinkage. Compare the response to halving and doubling the level and slope calibrations. Curves and bands show medians and pointwise 95% credible intervals. Joint innovation panels show how variation is allocated between level and slope. This figure is to be completed from validated fits.

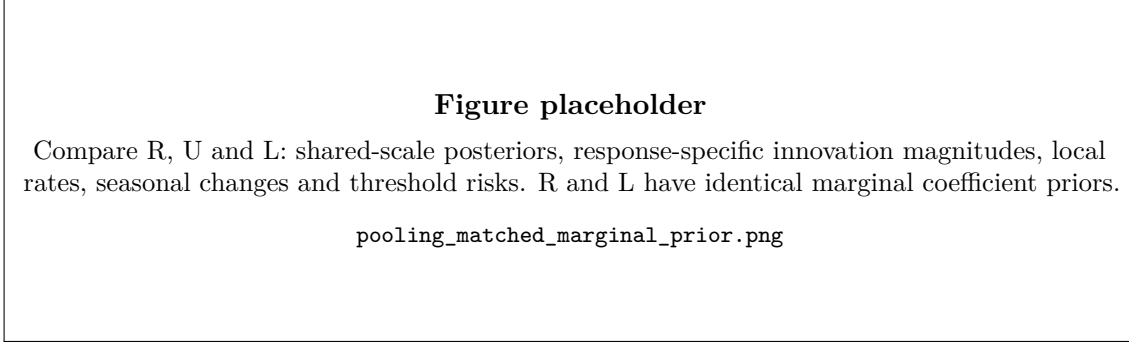


Figure S6: Information sharing and marginal prior shape. R uses common learned scales, U fixed normal priors with matched variances, and L independent half-normal scale mixtures with the same marginal priors as R. Report central 95% intervals for each response and keep the uncertainty in local scales visible.

S2.3 Half-normal, half- t and half-Cauchy hyperpriors

The half- t family provides a natural sensitivity comparison for hierarchical SDs (Gelman, 2006); the half-Cauchy has also been studied specifically for global shrinkage scales (Polson and Scott, 2012). All three families have positive density near zero. Their upper tails and the corresponding prior predictive distributions differ.

For H4, use a half- t with four degrees of freedom and scale $B_c = A_c/\sqrt{2}$. Because $\mathbb{E}(\tau_c^2) = \nu B_c^2/(\nu - 2)$ for $\nu > 2$, this choice preserves $\mathbb{E}(\tau_c^2) = A_c^2$, and hence the 30-year prior SDs. It changes the mixture’s shape without changing those second moments. The t scale parameter is not its standard deviation.

For C1, a half-Cauchy cannot be matched by variance because its second moment is infinite. Instead, match the 95th percentile of the shared scale to that of R:

$$\tau_c \sim \text{HalfCauchy}(B_c^C), \quad B_c^C = \frac{z_{0.975}}{t_{1,0.975}} A_c \simeq 0.1543 A_c. \quad (\text{S11})$$

Here z_u and $t_{\nu,u}$ are standard-normal and Student- t quantiles. Thus every family comparison has an explicit calibration; using the same numerical scale for all families would not accomplish this.

Matching this quantile also matches that of each contribution’s conditional prior SD, which is proportional to τ_c . It does not match the complete prior distribution of the temperature change. For C1 we report prior quantiles and tail probabilities, never a finite prior SD for the innovation contributions.

Table S3: Pooling and hyperprior-family comparisons. The initial-slope prior and observation model remain at the reference unless another perturbation is explicitly named.

Setting	Innovation prior	Calibration relative to R
R	Shared $\tau_c \sim \text{HalfNormal}(A_c)$	$\mathbb{E}(s_{c,j}^2) = A_c^2$
U	Fixed $s_{c,j} \sim \mathcal{N}(0, A_c^2)$	Same coefficient variances; no pooling
L	Independent $\tau_{c,j} \sim \text{HalfNormal}(A_c)$	Same marginal coefficient distributions; no pooling
H4	Shared $\tau_c \sim \text{Half-}t_4(A_c/\sqrt{2})$	Same coefficient variances and 30-year prior SDs
C1	Shared $\tau_c \sim \text{HalfCauchy}(0.1543A_c)$	Same 95th percentile of τ_c ; infinite second moment

For R, H4 and C1, we additionally halve and double all three anchors together while keeping P_{β_0} fixed. These checks separate the family comparison from uncertainty about the physical calibration. They do not assume that a heavier-tailed hyperprior must increase the posterior rate or improve predictive performance. Prior simulations show both near-zero behaviour and the tail; posterior comparisons examine trajectories and risks as well as the learned scales.

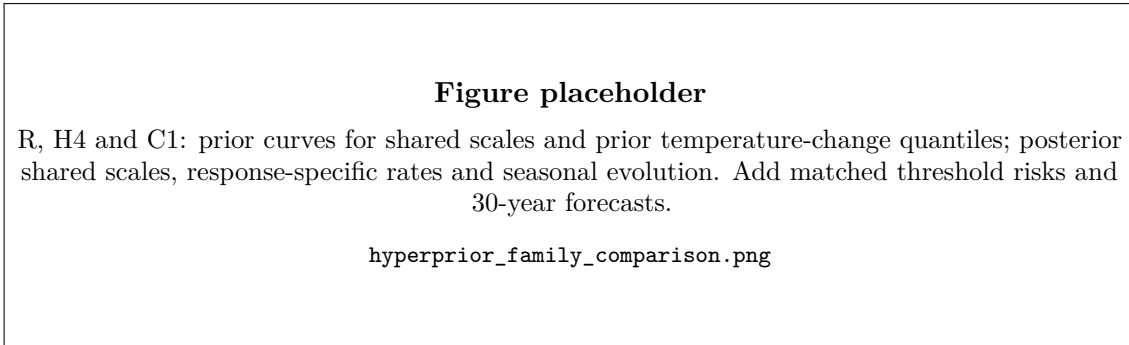


Figure S7: Sensitivity to the shrinkage hyperprior family. The half- t_4 comparison matches the reference second moments; the half-Cauchy comparison matches the 95th percentile of each shared scale. Display posterior medians and 95% credible intervals, together with the corresponding prior distributions. Do not report finite prior SDs for the half-Cauchy mixture.

S2.4 Computation and comparison quantities

The screening budget is two independent chains for each full model setting, with 700 warm-up and 1,300 retained iterations per chain. For pooled settings, each chain contains all six responses and the three shared scales; response-specific blocks can be parallelized conditional on those scales. U and L instead permit six independent marginal fits. The proposed longer runs use two chains with 2,000 warm-up and 4,000 retained iterations for alternatives, and 3,000 warm-up and 8,000 retained iterations for the reference. These are starting budgets, extended when needed to achieve adequate Monte Carlo precision.

Screen thresholds are $\hat{R} \leq 1.05$ and bulk and tail effective sample sizes of at least 100; the manuscript assessment uses $\hat{R} \leq 1.01$ and effective sample sizes of at least 400 for the reported targets (Vehtari et al., 2021). We inspect the shared scales, physical coefficient magnitudes, initial and terminal rates, state summaries and event probabilities. Numerical failures remain visible. Apparent stability from poorly mixing chains is not evidence of robustness.

Comparisons use the same observations and original temperature orientation. We examine $|s_{\alpha,j}|$, $|s_{\beta,j}|$, $|s_{\gamma,j}|$, $40\beta_{j,0}$, the full level and rate trajectories, and seasonal evolution. Conditional 10- and 30-year innovation contributions are computed from the fitted $|s_{c,j}|$, using the gains in [appendix S1.4](#). Prior-to-posterior changes indicate learning but are not a criterion for selecting a preferred prior.

Risk comparisons retain JJA TXx above 35 °C, JJA TNx above 25 °C, DJF TXn below 0 °C and DJF TNn below −10 °C. All full-record settings use a common 30-year forecast horizon. The pre-2019 analysis compares R, U, H4, C1, S14, S15 and S18, fitting only observations through May 2019 and evaluating TXx exceedances of 35 °C, 36.6 °C and 39.7 °C. Every pooled pre-2019 fit uses only pre-2019 observations of all six responses; full-record shared scales must not be reused.

S2.5 Figures and findings to be completed

Sensitivity findings pending

The numerical Results section is retained from the supplied draft and has not been recomputed under the half-normal hierarchy. Replace the placeholders below after checking the new fits. Report whether pooling actually reduces sensitivity of rates and risks, whether seasonal evolution remains identifiable, and which conclusions change under H4 or C1. The intended robustness benefit is not an established result of this revised hierarchy until these comparisons are available.

Figure placeholder

Three shared scales, then all six responses: physical level, slope and seasonal innovation magnitudes and initial rates. Show marginal priors integrated over the hyperpriors and the corresponding posteriors.

`sensitivity_prior_posterior.png`

Figure S8: Prior-to-posterior learning under hierarchical shrinkage. Separate learning about the common prior scales from learning about each response’s actual innovation magnitudes. Initial rates use their fixed normal prior and original temperature orientation. All intervals are 95%.

Figure placeholder

Levels and rates for all six summaries under R and S1–S8. Separate simultaneous calibration changes from single-component changes.

`sensitivity_structural_trajectories.png`

Figure S9: Sensitivity of levels and rates to the hierarchical calibration. Compare reference and half/double anchor settings using posterior medians and pointwise 95% credible bands on common axes. The shared scales are re-estimated in every fit.

Figure placeholder

R, S9 and S10: initial-rate prior/posterior distributions, complete rate trajectories, and terminal-rate estimates with 95% intervals.

`sensitivity_initial_rates.png`

Figure S10: Sensitivity to the separately calibrated initial-rate prior. The prior SDs are 0.20, 0.40 and 0.80 degrees Celsius per decade. The three innovation shrinkage scales remain shared and learned.

Figure placeholder

R, S7, S8, S11–S13 and S21: same-season components, warming rates, and conditional 10-/30-year seasonal innovation contributions.

`sensitivity_seasonal_evolution.png`

Figure S11: Seasonal shrinkage and allocation of persistent change. Compare seasonal calibration and doubled-slope combinations with the fixed-seasonality alternative. Display posterior medians and 95% intervals, including the fixed model's zero innovation contribution.

Figure placeholder

R versus S18–S20 and S22: seasonal observation scales, initial seasonal coefficients, and changes in fitted levels and rates.

`sensitivity_scale_initial_cycle.png`

Figure S12: Observation dispersion and the initial seasonal pattern. Compare constant dispersion, seasonal log-scale prior widths and the tighter initial-cycle prior using central 95% intervals.

Figure placeholder

R versus S14–S17: GEV shape posteriors and historical threshold risks. Additional panels compare 30-year forecasts under structural perturbations.

`sensitivity_shape_risk.png`

Figure S13: Sensitivity of tail inference and evolving risk. Display the four shape posteriors and conditional event probabilities, with pointwise 95% intervals and original cold-tail orientation. Record any indirect changes in the Gaussian responses through pooling.

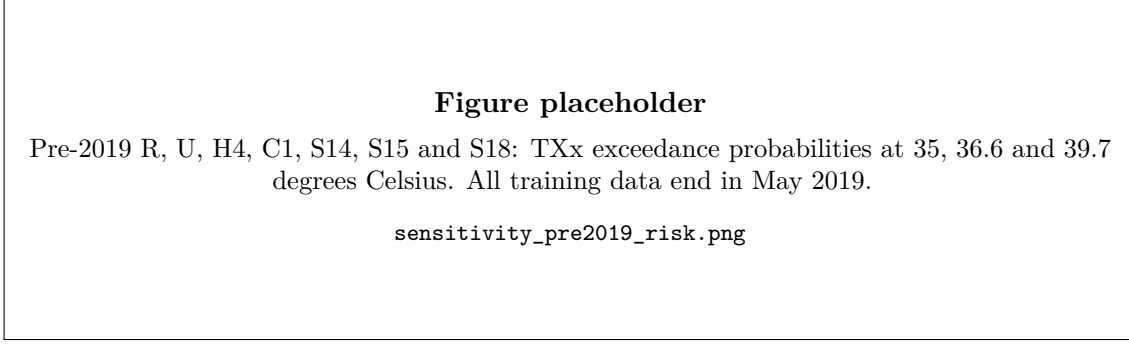


Figure S14: Sensitivity of the prospective 2019 heat-risk assessment. Points are posterior predictive probabilities; intervals are 95% intervals for conditional risks. All shared scales are learned using only the pre-event training record.

S3 Conditional likelihoods and posterior updates

S3.1 Posterior target

Let $\tilde{\mathbf{x}}_j$ denote the standardized paths, $\theta_j = (\alpha_{j,0}, \beta_{j,0}, \gamma_j^{(0)}, \mathbf{s}_j)$ the structural coefficients, and $\eta_j = (\kappa_j, \mathbf{u}_j, \xi_j)$ the observation parameters, omitting shape for Gaussian responses. With $\mathbf{Y} = (\mathbf{Y}_1, \dots, \mathbf{Y}_6)$, the joint target is

$$\begin{aligned} \pi(\boldsymbol{\tau}, \{\tilde{\mathbf{x}}_j, \theta_j, \eta_j\}_{j=1}^6 \mid \mathbf{Y}) &\propto p(\boldsymbol{\tau}) \prod_{j=1}^6 p_0(\tilde{\mathbf{x}}_j) p(\theta_j \mid \boldsymbol{\tau}) p(\eta_j) \\ &\times \prod_{j=1}^6 \prod_{t=1}^T g_{j,t}(Y_{j,t} \mid \tilde{\mathbf{x}}_j, \theta_j, \eta_j). \end{aligned} \quad (\text{S12})$$

The fixed A_c enter $p(\boldsymbol{\tau})$; P_{β_0} enters the initial-slope normal prior. Conditional on $\boldsymbol{\tau}$, each response-specific block uses its own observations. Updating the common scales then transfers information among those blocks. Independent chains must target this complete hierarchy, rather than fitting the six responses once and pooling their posterior estimates afterwards.

S3.2 Gaussian state update

For a Gaussian response, subtract the deterministic part of [equation \(5\)](#) to obtain

$$\begin{aligned} Y_{j,t} - \alpha_{j,0} - t\beta_{j,0} - \gamma_{j,s(t)}^{(0)} &= s_{\alpha,j}\tilde{\alpha}_{j,t} + s_{\beta,j}\tilde{A}_{j,t} + s_{\gamma,j}\tilde{\gamma}_{j,t} + e_{j,t}, \\ e_{j,t} &\sim \mathcal{N}(0, \sigma_{j,t}^2). \end{aligned} \quad (\text{S13})$$

Conditional on the coefficients and observation scales, this is a linear Gaussian observation equation for standardized states with Gaussian transitions. Kalman filtering followed by backward simulation samples the complete path from its exact conditional posterior ([Carter and Kohn, 1994](#); [Frühwirth-Schnatter, 1994](#); [Durbin and Koopman, 2000](#)).

S3.3 GEV path proposal and exact correction

For computation, a lower extreme is reflected, $W_{j,t} = -Y_{j,t}$, and evaluated with the maximum-GEV kernel using location $-\mu_{j,t}$. This is algebraically equivalent to [equation \(8\)](#); all reported quantities are transformed back to their original orientation.

For the maximum-GEV kernel, let $h = 1 + \xi(w - \mu)/\sigma > 0$. Its log density and first two location derivatives are

$$\begin{aligned}\log f(w) &= -\log \sigma - (1 + 1/\xi) \log h - h^{-1/\xi}, \\ \partial_\mu \log f(w) &= \frac{1 + \xi}{\sigma h} - \frac{h^{-1/\xi-1}}{\sigma}, \\ \partial_\mu^2 \log f(w) &= \frac{\xi(1 + \xi)}{\sigma^2 h^2} - \frac{(1 + \xi)h^{-1/\xi-2}}{\sigma^2}.\end{aligned}\tag{S14}$$

Continuous limits are used near $\xi = 0$. A deterministic mode and curvature calculation based on the GEV likelihood and Gaussian state prior defines the Gaussian approximation $\tilde{L}$, which remains fixed throughout the path update.

The exact conditional target and the proposal are proportional to

$$p_0(\tilde{\mathbf{x}}_j)L(\tilde{\mathbf{x}}_j) \quad \text{and} \quad p_0(\tilde{\mathbf{x}}_j)\tilde{L}(\tilde{\mathbf{x}}_j),$$

respectively. Substitution into the independence-MH ratio cancels the Gaussian state prior and gives [equation \(16\)](#). The exact likelihood L retains the GEV density and support. Non-quadratic behavior near a finite endpoint can therefore lower acceptance but cannot replace the posterior target by the approximation.

S3.4 Remaining parameter updates

Conditional on the paths, Gaussian structural coefficients are sampled from their Gaussian full conditionals. For a GEV coefficient block, write its exact conditional target as

$$\pi(\theta) = p_G(\theta) \frac{\pi(\theta)}{p_G(\theta)},$$

where p_G is a fixed, full-support Gaussian reference obtained from a deterministic local mode calculation. Elliptical slice sampling uses $\log \pi - \log p_G$ as its residual log likelihood and therefore retains the exact conditional target ([Murray et al., 2010](#)).

The three seasonal log-scale coordinates for a response are updated jointly by elliptical slice sampling. Baseline log scales and GEV shapes are updated componentwise by stepping-out slice sampling ([Neal, 2003](#)). Updates on transformed scales include the corresponding prior and Jacobian. For example, if $\sigma^2 \sim IG(a, b)$ and $\ell = \log \sigma$, the log prior in ℓ is $-2a\ell - b \exp(-2\ell)$, up to a constant. The innovation coefficients and initial slope are updated within the structural block conditional on the current $\boldsymbol{\tau}$. The initial-slope variance is fixed, while the three shared scales are updated as follows.

S3.5 Shared shrinkage update

Write $S_c = \sum_{j=1}^K s_{c,j}^2$, with $K = 6$. For the half-normal reference, the full conditional is

$$p(\tau_c \mid \{s_{c,j}\}, A_c) \propto \tau_c^{-K} \exp\left\{-\frac{S_c}{2\tau_c^2} - \frac{\tau_c^2}{2A_c^2}\right\}, \quad \tau_c > 0.\tag{S15}$$

Updating $\ell_c = \log \tau_c$ by stepping-out slice sampling uses

$$\log p(\ell_c \mid \cdots) = -(K-1)\ell_c - \frac{S_c}{2}e^{-2\ell_c} - \frac{e^{2\ell_c}}{2A_c^2} + \text{constant}.\tag{S16}$$

The $+\ell_c$ Jacobian is included. The three updates are independent conditional on the coefficient sets. This formula also makes the source of pooling explicit: every response contributes its squared coefficient to the same S_c .

For a half- t_ν scale prior with scale B_c , including the half-Cauchy at $\nu = 1$, replace the final exponential penalty by

$$-\frac{\nu+1}{2} \log \left(1 + \frac{e^{2\ell_c}}{\nu B_c^2} \right).$$

The likelihood term and Jacobian are unchanged. A sweep updates each response's path, coefficients and observation parameters, then the three shared scales. Conditional response updates and independent chains can be parallelized, but the shared-scale block must use all six current coefficient sets.

S4 Focused adequacy and predictive assessment

S4.1 Marginal fit and numerical diagnostics

The reference, constant-dispersion and fixed-seasonality specifications are R, S18 and S21 in [table S2](#). Their comparison assesses seasonal observation variability and evolution of the seasonal location pattern while retaining the same reference level and slope priors. The corresponding sensitivity displays are specified in [appendix S2](#).

Posterior predictive checks compare observed and replicated seasonal spread, tails, threshold counts and residual serial structure ([Gelman et al., 1996](#)). Smoothed PIT values reuse the fitted observations; they are descriptive diagnostics rather than calibrated one-step predictive uniforms. Observed and replicated discrepancies are computed under the same posterior draw.

Convergence is assessed for the shared shrinkage scales, physical innovation magnitudes, initial rates, state summaries, observation parameters and risk probabilities. Switching between equivalent signs of an innovation coefficient does not demonstrate adequate mixing of its magnitude. Numerical flags remain visible in all comparisons; satisfactory MCMC diagnostics do not establish observation-model adequacy.

Figure to insert: `sensitivity_seasonal_pit_serial.png`

Figure S15: Seasonal marginal and serial diagnostics. Placeholder for season-specific PIT and normal-score Q-Q plots, residual autocorrelations and matched posterior predictive checks. Compare the reference with the scale or seasonality alternatives that address the observed discrepancies.

S4.2 Held-out predictive assessment

Expanding-window forecasts use only observations available at each origin ([Bürkner et al., 2020](#)). The reference five-year validation uses origins at the end of SON in 1970, 1980, 1990, 2000, 2010, 2015 and 2020. The sensitivity validation compares R with U, H4, C1, S1–S15, S18 and S21 at the common 2010, 2015 and 2020 origins. For every pooled setting, all six training series stop at the same origin and the shared scales are re-estimated without access to held-out observations. The original split design additionally uses the first 60%, 80% and 90% of the record for training and a ten-year forecast horizon under R. These held-out runs are separate from the full-record sensitivity fits.

Comparisons pair the same response, forecast origin, season and lead. Report CRPS with central 90%, 95% and 99% coverage and interval widths, and inspect predictive bias and directional tail

misses. A central 99% interval assesses the 0.5% and 99.5% quantiles, not the upper 99% quantile. The number of forecast cases and threshold events is reported by season; incomplete origin groups are not ranked. Finite held-out samples cannot establish arbitrary far-tail calibration.

Held-out results pending

Fill the following figure after the separate validation runs are available. Full-record screen fits alone do not provide held-out CRPS, coverage or predictive PIT evidence.

Figure to insert: `sensitivity_heldout_scores_calibration.png`

Figure S16: Held-out predictive sensitivity. Placeholder for paired CRPS comparisons, 90%/95%/99% coverage and widths, predictive bias and PIT plots by response, season and forecast origin. Identify numerically flagged fits and show case counts.

S5 Forecast propagation and aggregation

S5.1 Conditional probabilities and return levels

For each posterior draw, the original-temperature CDF $G_{j,t}$ gives the threshold probabilities in [equation \(20\)](#). Stationary-equivalent return periods are computed by taking the reciprocal within each draw, and return levels by inverting its CDF. Their posterior medians and central 95% intervals describe uncertainty about these conditional quantities.

For a maximum-GEV margin, the conditional quantile at probability $u \in (0, 1)$ is

$$Q^+(u; \mu, \sigma, \xi) = \begin{cases} \mu + \frac{\sigma}{\xi} \{(-\log u)^{-\xi} - 1\}, & \xi \neq 0, \\ \mu - \sigma \log(-\log u), & \xi = 0. \end{cases} \quad (\text{S17})$$

The minimum-GEV quantile in the parameterization of [equation \(8\)](#) is $Q^-(u; \mu, \sigma, \xi) = -Q^+(1 - u; -\mu, \sigma, \xi)$. For Gaussian margins it is $\mu + \sigma \Phi^{-1}(u)$. An upper-tail r -year return level for a specified season uses $u = 1 - 1/r$, and a lower-tail level uses $u = 1/r$. The fitted CDF includes the appropriate GEV endpoint: a threshold outside the attainable tail can have conditional event probability zero and hence an infinite stationary-equivalent return period.

S5.2 Propagation of forecast uncertainty

Let $\mathcal{F}_T$ denote the observed record. For posterior draw k , retain the terminal state and parameters, then simulate

$$\mathbf{x}_{j,T+h}^{(k)} = F \mathbf{x}_{j,T+h-1}^{(k)} + L_j^{(k)} \boldsymbol{\varepsilon}_{j,T+h}^{(k)}, \quad h = 1, \dots, h_{\max}, \quad (\text{S18})$$

using independent standard-normal innovations. The matrices are defined in [appendix S1.1](#). A 30-year forecast contains $h_{\max} = 120$ seasonal transitions.

Conditional on the terminal state and parameters, write the mean state forecast as $\mathbf{x}_{j,T+h|T} = F^h \mathbf{x}_{j,T}$. Its first seasonal coordinate is $\gamma_{j,T+h|T}$ in [equation \(19\)](#). It is obtained recursively from

$$\gamma_{j,T+h|T} = - \sum_{r=1}^3 \gamma_{j,T+h-r|T}, \quad h \geq 1, \quad (\text{S19})$$

starting with $\gamma_{j,T-r|T} = \gamma_{j,T-r}$ for $r = 0, 1, 2$. The first four forecasts are therefore

$$-\gamma_{j,T} - \gamma_{j,T-1} - \gamma_{j,T-2}, \quad \gamma_{j,T-2}, \quad \gamma_{j,T-1}, \quad \gamma_{j,T}.$$

This forecast cycle repeats every four steps. Individual simulated cycles continue to evolve through future seasonal innovations.

Writing $Q_j = L_j L_j^\top$, the conditional state covariance is

$$\text{Var}(\mathbf{x}_{j,T+h} \mid \mathbf{x}_{j,T}, \Theta_j) = \sum_{r=0}^{h-1} F^r Q_j (F^r)^\top. \quad (\text{S20})$$

Multiplication by H and $H^\top$ gives the conditional location variance. In particular, the trend contribution is $h s_{\alpha,j}^2 + s_{\beta,j}^2 h(h-1)(2h-1)/6$, as in [equation \(22\)](#). Simulation across posterior draws also integrates uncertainty in the terminal state and parameters.

For each future state, drawing an observation from $G_{j,T+h}$ adds weather variability and gives the posterior predictive distribution. Its 2.5% and 97.5% quantiles form a 95% prediction interval. Alternatively, averaging the conditional CDFs gives

$$G_{j,T+h}^{\text{pred}}(c) = \mathbb{E}\{G_{j,T+h}(c) \mid \mathcal{F}_T\}, \quad (\text{S21})$$

without additional observation simulation. Its tail probabilities are forecast event probabilities. Quantiles of this mixture are distinct from posterior averages or medians of conditional quantiles: return-level forecasts with credible intervals are obtained by calculating the conditional quantile along each future trajectory, whereas inverting $G_{j,T+h}^{\text{pred}}$ gives a predictive quantile that integrates over those trajectories. Predictive quantiles remain defined even when some GEV draws have infinite observation moments.

S5.3 Annual aggregation from seasonal blocks

Let $\mathcal{S}_y$ contain DJF, MAM, JJA and SON labelled by year y . Their union is the December–November annual block, since DJF is labelled by the year in which it ends. For a maximum response j , write $X_{j,d}$ for its underlying daily temperature (TX or TN). The annual maximum satisfies the exact identity

$$M_{j,y} = \max_{t \in \mathcal{S}_y} Y_{j,t} = \max_{t \in \mathcal{S}_y} \max_{d \in B_t} X_{j,d} = \max_{d \in \bigcup_{t \in \mathcal{S}_y} B_t} X_{j,d}. \quad (\text{S22})$$

Replacing maxima by minima gives the corresponding identity for an annual minimum. These identities require no independence assumption and introduce no additional extreme-value approximation. A January–December annual extreme cannot generally be reconstructed from these four seasonal extrema, because DJF crosses the calendar-year boundary.

The distributional calculation uses the model’s conditional independence of observations across seasons. Given the complete latent path and parameters, the annual maximum and minimum CDFs are, respectively,

$$F_{j,y}^{\text{ann},+}(c) = \prod_{t \in \mathcal{S}_y} G_{j,t}(c), \quad F_{j,y}^{\text{ann},-}(c) = 1 - \prod_{t \in \mathcal{S}_y} \{1 - G_{j,t}(c)\}. \quad (\text{S23})$$

The resulting annual distribution need not itself be GEV. Annual conditional threshold probabilities and return levels follow by evaluating or inverting these CDFs within each posterior draw.

For a complete future annual block, integrating over the parameters and future latent trajectory gives

$$\begin{aligned} \mathbb{P}(M_{j,y} \leq c \mid \mathcal{F}_T) &= \mathbb{E} \left[\prod_{t \in \mathcal{S}_y} G_{j,t}(c) \mid \mathcal{F}_T \right], \\ \mathbb{P}(L_{j,y} \leq c \mid \mathcal{F}_T) &= 1 - \mathbb{E} \left[\prod_{t \in \mathcal{S}_y} \{1 - G_{j,t}(c)\} \mid \mathcal{F}_T \right], \end{aligned} \quad (\text{S24})$$

where $M_{j,y}$ applies to a maximum response and $L_{j,y}$ to a minimum response. Each product is evaluated along one simulated latent trajectory before averaging. The product is formed before posterior averaging because its factors share uncertain parameters and latent states. In general, an expectation of this product cannot be replaced by a product of its expectations.

Inverting the posterior-averaged CDF in [equation \(S24\)](#) gives a quantile of the annual predictive distribution, rather than a posterior summary of conditional return levels. For the two mean responses, the annual daily mean is recovered by weighting the seasonal means by their block lengths:

$$\bar{Y}_{j,y} = \frac{\sum_{t \in \mathcal{S}_y} |B_t| Y_{j,t}}{\sum_{t \in \mathcal{S}_y} |B_t|}. \quad (\text{S25})$$

Historical annual summaries use complete blocks. Forecasts of an annual block already partly observed retain its observed seasonal values and simulate only the remaining blocks.

S5.4 Risk over a specified future period

For a collection of future blocks $\mathcal{H}$, the probability of at least one upper-threshold exceedance is

$$\mathbb{P}\left(\max_{t \in \mathcal{H}} Y_{j,t} > c \mid \mathcal{F}_T\right) = 1 - \mathbb{E}\left\{\prod_{t \in \mathcal{H}} G_{j,t}(c) \mid \mathcal{F}_T\right\}. \quad (\text{S26})$$

For at least one lower-threshold crossing, replace each $G_{j,t}(c)$ by $1 - G_{j,t}(c)$ inside the product. These calculations allow risks to evolve over the specified period ([Rootzén and Katz, 2013](#)).

Conditional-CDF calculations are especially useful for very rare events, whose probabilities would be estimated imprecisely from a small number of binary exceedance indicators. Averaging conditional probabilities gives the posterior predictive event probability; their 2.5% and 97.5% quantiles describe uncertainty in the conditional risk over parameters and latent trajectories. Quantiles of binary indicators do not give such an interval. Monte Carlo error is assessed separately. Agreement between annual simulation and [equation \(S24\)](#) checks the aggregation calculation, rather than validating conditional independence between seasons.